

Matrices

Unit 2-3 Lecture Notes

MSDS 520: Math Fundamentals for Data Science
Meharry Medical College

These notes are the written companion to the Unit 2-3 slides and follow them section by section. Every exercise is followed by a worked solution; try each one before reading on.

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1 Where Vectors Left Off

1.1 A Dependence Problem, Four Ways

Unit 2-2 ended with a question about redundancy. Take the three vectors

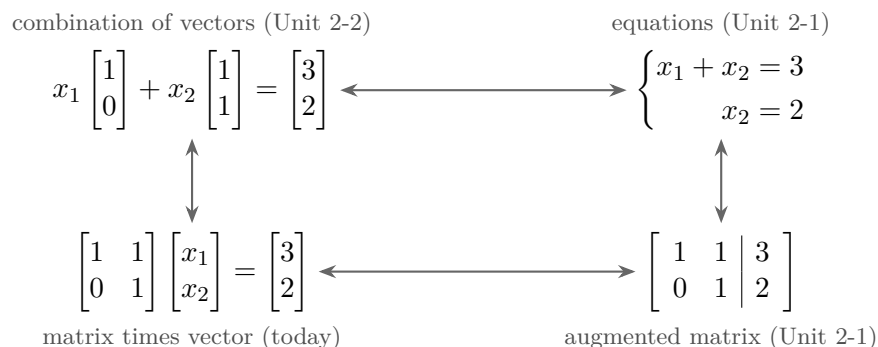
$$\bar{u} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \quad \bar{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \bar{w} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Is $\bar{u}$ a **combination** of $\bar{v}$ and $\bar{w}$? That is, can we reach $\bar{u}$ by scaling $\bar{v}$ and $\bar{w}$ and adding the results? If we can, then $\bar{u}$ is **redundant**: it adds no direction that $\bar{v}$ and $\bar{w}$ do not already supply.

Why would anyone care? In a dataset, a redundant vector is a feature (or a record) that carries no new information: a column that can be rebuilt exactly from other columns. Unit 2-2 saw the classic case, a patient's height recorded once in inches and again in centimeters. Spotting redundancy is

the first step in cleaning such data, and the question “is this vector a combination of those?” is how you spot it.

By now we can write this one question in four different ways, and it pays to see them side by side. Each form in the diagram says exactly the same thing; the double arrows mean “says the same thing as.”



Walk around the diagram clockwise from the top left.

- **Combination of vectors.** This is the question as Unit 2-2 posed it: find amounts x_1 and x_2 of $\bar{v}$ and $\bar{w}$ that add up to $\bar{u}$.
- **Equations.** Two vectors are equal when their entries agree, so compare the two sides one entry at a time. The top entries give $x_1 + x_2 = 3$, and the bottom entries give $0x_1 + 1x_2 = 2$, which is just $x_2 = 2$. This is a system of equations, exactly as in Unit 2-1.
- **Augmented matrix.** Unit 2-1 kept only the numbers: each row is one equation, and the bar separates the coefficients from the right-hand sides.
- **Matrix times vector.** This is the new form, and the subject of today. Section 5 defines the product properly; for now, read it as shorthand for the combination at the top left.

Notice where $\bar{v}$ and $\bar{w}$ ended up. In the augmented matrix and in the new form alike, they stand side by side as the **columns** of a grid: the first column is $\bar{v}$ and the second is $\bar{w}$. That observation is the key to everything in these notes.

Naming the pieces: $A\bar{x} = \bar{b}$

- $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ is the **coefficient matrix**. Its columns are $\bar{v}$ and $\bar{w}$.
- $\bar{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ holds the **unknowns**: how much of each column to use.
- $\bar{b} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ is the **target**: the right-hand side, here $\bar{u}$.

$A\bar{x} = \bar{b}$ reads: **weight the columns of A by $\bar{x}$, and hit $\bar{b}$.**

The short form has one big advantage: it looks the same no matter how large the problem is. Two vectors in the plane or many vectors with many entries, the question is still written $A\bar{x} = \bar{b}$.

Watch out: the vectors go in as columns, not rows

The most common slip when setting up A is to lay the vectors in as **rows**. With $\bar{v}$ and $\bar{w}$ as rows, the grid would be $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, whose equations are $x_1 = 3$ and $x_1 + x_2 = 2$, with solution $x_1 = 3$, $x_2 = -1$. But $3\bar{v} - \bar{w} = \begin{bmatrix} 3 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, which is not $\bar{u}$. The rows version answers a different question. Each unknown multiplies a whole **vector**, so each vector must occupy the slot that one unknown multiplies: a column.

Example 1.1: The four forms for a new pair of vectors. Is $\bar{b} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$ a combination of $\bar{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\bar{w} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$? Write the question all four ways first; the columns of the coefficient matrix are $\bar{v}$ and $\bar{w}$, in that order.

$$x_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix} \quad \longleftrightarrow \quad \begin{cases} 2x_1 + x_2 = 5 \\ x_1 + 3x_2 = 5 \end{cases}$$

$$\longleftrightarrow \quad \left[\begin{array}{cc|c} 2 & 1 & 5 \\ 1 & 3 & 5 \end{array} \right] \quad \longleftrightarrow \quad \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$$

Read the equations off the combination one entry at a time: the top entries give $2x_1 + x_2 = 5$ and the bottom entries give $x_1 + 3x_2 = 5$. Neither equation hands us an unknown directly, so use substitution from Unit 2-1. The second equation gives $x_1 = 5 - 3x_2$; putting that into the first,

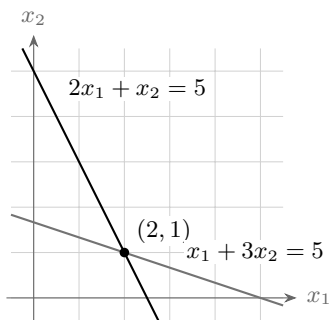
$$2(5 - 3x_2) + x_2 = 5 \implies 10 - 5x_2 = 5 \implies x_2 = 1,$$

and then $x_1 = 5 - 3(1) = 2$. Check it as a combination:

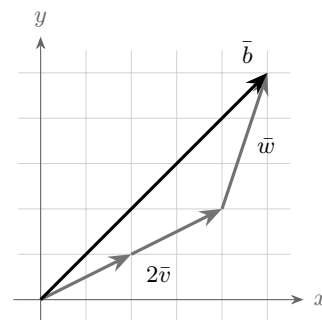
$$2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 + 1 \\ 2 + 3 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}. \checkmark$$

So $\bar{b} = 2\bar{v} + \bar{w}$: the recipe is “two of the first column, one of the second.”

Unit 2-2 gave two pictures of a system like this one, and both are worth drawing. In the **row picture** each equation is a line in the (x_1, x_2) plane, and the solution is the point where the lines cross. In the **column picture** the unknowns are amounts of the two column vectors, laid head to tail until they reach the target.



Row picture: the lines cross at $(2, 1)$.



Column picture: $2\bar{v}$, then $\bar{w}$, lands on $\bar{b}$.

The two pictures answer the same question from opposite directions. The crossing point $(2, 1)$ in the row picture is the *list of amounts*, and the column picture shows those amounts at work.

Example 1.2: A supplement mix, with more entries than unknowns. A clinic dietitian combines two protein supplements. One serving of supplement 1 supplies 10 g of protein, 5 g of carbohydrate, and 2 g of fat; one serving of supplement 2 supplies 5 g, 10 g, and 1 g. Can some number of servings supply exactly 25 g of protein, 20 g of carbohydrate, and 5 g of fat? Each supplement is a vector with one entry per nutrient, and the question is a combination:

$$x_1 \begin{bmatrix} 10 \\ 5 \\ 2 \end{bmatrix} + x_2 \begin{bmatrix} 5 \\ 10 \\ 1 \end{bmatrix} = \begin{bmatrix} 25 \\ 20 \\ 5 \end{bmatrix} \quad \longleftrightarrow \quad \underbrace{\begin{bmatrix} 10 & 5 \\ 5 & 10 \\ 2 & 1 \end{bmatrix}}_A \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 25 \\ 20 \\ 5 \end{bmatrix}.$$

This coefficient matrix has three rows (one per nutrient) and two columns (one per supplement); it need not be square. Comparing entries gives three equations in two unknowns:

$$\begin{cases} 10x_1 + 5x_2 = 25 \\ 5x_1 + 10x_2 = 20 \\ 2x_1 + x_2 = 5 \end{cases}$$

Dividing the first equation by 5 gives $2x_1 + x_2 = 5$, which is the third equation again, so only two different facts remain. From $2x_1 + x_2 = 5$ we get $x_2 = 5 - 2x_1$, and the second equation (divided by 5) reads $x_1 + 2x_2 = 4$, so

$$x_1 + 2(5 - 2x_1) = 4 \implies 10 - 3x_1 = 4 \implies x_1 = 2, \quad x_2 = 5 - 4 = 1.$$

Check it as a combination:

$$2 \begin{bmatrix} 10 \\ 5 \\ 2 \end{bmatrix} + 1 \begin{bmatrix} 5 \\ 10 \\ 1 \end{bmatrix} = \begin{bmatrix} 20 + 5 \\ 10 + 10 \\ 4 + 1 \end{bmatrix} = \begin{bmatrix} 25 \\ 20 \\ 5 \end{bmatrix}. \checkmark$$

Two servings of supplement 1 and one of supplement 2 hit the target exactly. With three equations and only two unknowns, that is a small piece of luck, and the next subsection shows how easily it breaks.

1.2 Solving It: One Equation at a Time

For a problem this small, the equations are the easiest form to solve by hand. Read them from the bottom up; each one hands you an unknown:

$$\begin{cases} x_1 + x_2 = 3 \\ x_2 = 2 \end{cases} \implies x_2 = 2, \quad x_1 = 3 - x_2 = 1.$$

The bottom equation mentions only x_2 , so it gives $x_2 = 2$ at once. Putting that into the top equation leaves $x_1 + 2 = 3$, so $x_1 = 1$. This is the back-substitution of Unit 2-1.

Read it off. The solution is $\bar{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$: one of the first column and two of the second. In terms of the original vectors, $\bar{u} = \bar{v} + 2\bar{w}$. Check it:

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 + 2 \\ 0 + 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

So $\bar{u}$ is redundant, and the set $\{\bar{u}, \bar{v}, \bar{w}\}$ is **dependent**. Notice that the solution does more than say “yes”: the solution **is** the recipe. Its entries tell you exactly how much of each column to use to build $\bar{u}$.

Change one number. Now put $\bar{w}$ on $\bar{v}$ ’s line instead, say $\bar{w} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$. The same question becomes

$$x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \quad \longleftrightarrow \quad \begin{cases} x_1 + 2x_2 = 3 \\ 0 = 2 \end{cases}$$

The second equation now reads $0 = 2$, which is false whatever x_1 and x_2 are, so there is **no solution**. The geometry says the same thing. Both $\bar{v}$ and $\bar{w}$ now lie along the x -axis, so every combination of them does too: they span only the x -axis, and $\bar{u}$ is off it. An impossible equation like $0 = 2$ is the algebra’s way of saying the target is out of reach.

Change the target as well. Keep $\bar{w} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$, but aim at a target that *is* on the x -axis, $\begin{bmatrix} 3 \\ 0 \end{bmatrix}$. The equations become $x_1 + 2x_2 = 3$ and $0 = 0$. The second is always true and tells us nothing, so the first is the only condition, and it has infinitely many solutions: pick any x_2 and set $x_1 = 3 - 2x_2$. For instance

$$\bar{x} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}, \quad \bar{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \bar{x} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

all work (check the middle one: $1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$). Many recipes for the same target is the signal that the *columns themselves* are redundant: here $\bar{w} = 2\bar{v}$, so any amount of $\bar{w}$ can be traded for twice that amount of $\bar{v}$.

These are exactly the three outcomes of Unit 2-1, now read as statements about vectors.

The system has	In vector language
exactly one solution	the target is reachable, by exactly one recipe
no solution (a row like $0 = 2$)	the target is outside the span of the columns
infinitely many solutions	the target is reachable, and the columns are themselves dependent

Example 1.3: The supplement mix, one gram off. Keep the two supplements of Example 1.2, but change the fat target from 5 g to 6 g. The equations are now

$$\begin{cases} 10x_1 + 5x_2 = 25 \\ 5x_1 + 10x_2 = 20 \\ 2x_1 + x_2 = 6 \end{cases}$$

Divide the first by 5: it says $2x_1 + x_2 = 5$. The third says $2x_1 + x_2 = 6$. The same quantity cannot equal both 5 and 6; subtracting one from the other gives $0 = 1$. There is **no solution**, so no mix of these two supplements hits all three targets. Geometrically, the two supplement vectors span only a plane inside the three-dimensional space of nutrient totals, and the new target sits off that plane. With more targets than ingredients, “no exact recipe” is the usual state of affairs, which is why fitting problems in data science ask for the *closest* combination instead (least squares, which Section 6 touches on through $X^T X$).

1.3 Extra Practice

Exercise 1.1

Is $\bar{b} = \begin{bmatrix} 1 \\ 8 \end{bmatrix}$ a combination of $\bar{v} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\bar{w} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$? Write the question as a combination, as equations, and with the coefficient matrix named, then solve it and check your answer.

Solution to Exercise 1.1

The coefficient matrix has $\bar{v}$ and $\bar{w}$ as its columns:

$$x_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + x_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 8 \end{bmatrix} \quad \begin{cases} x_1 - x_2 = 1 \\ 2x_1 + x_2 = 8 \end{cases} \quad A = \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}.$$

Elimination is quickest: adding the two equations cancels x_2 and leaves $3x_1 = 9$, so $x_1 = 3$. Then the first equation gives $x_2 = x_1 - 1 = 2$. Check:

$$3 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 - 2 \\ 6 + 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 8 \end{bmatrix}. \checkmark$$

Yes: $\bar{b} = 3\bar{v} + 2\bar{w}$, so $\{\bar{b}, \bar{v}, \bar{w}\}$ is dependent and the recipe is $\bar{x} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$.

Exercise 1.2

Let $\bar{v} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $\bar{w} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$.

1. Is $\begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix}$ a combination of $\bar{v}$ and $\bar{w}$?
2. Change only the last entry of that target so that the answer becomes yes, and give the recipe.

Solution to Exercise 1.2

1. Comparing entries of $x_1\bar{v} + x_2\bar{w}$ with the target gives

$$\begin{cases} x_1 = 2 \\ x_1 + x_2 = 5 \\ x_2 = 1 \end{cases}$$

The first equation hands us $x_1 = 2$ and the third hands us $x_2 = 1$. But then the middle equation says $2 + 1 = 5$, which is false. The first and third equations have used up all our freedom, and the middle one disagrees, so there is **no solution**: the target is out of reach.

2. Keep $x_1 = 2$ from the first entry. The middle equation then needs $x_2 = 5 - 2 = 3$, so the last entry, which equals x_2 , must be 3. The target $\begin{bmatrix} 2 \\ 5 \\ 3 \end{bmatrix}$ works with recipe $x_1 = 2, x_2 = 3$: indeed

$$2\bar{v} + 3\bar{w} = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 3 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 3 \end{bmatrix}.$$

In words: two vectors in $\mathbb{R}^3$ span only a plane, and only targets on that plane can be reached. Changing one entry moved the target onto the plane.

2 Why Matrices?

2.1 A Problem to Motivate Us

Recall the matinee from Unit 2-1:

“A theater sold 300 tickets: children’s at \$6, adults’ at \$10, seniors’ at \$8, and took in \$2540. As many adult tickets were sold as children’s and seniors’ combined. How many of each?”

Let c , a , and s be the numbers of children’s, adults’, and seniors’ tickets. The problem gives three facts: 300 tickets were sold, \$2540 was taken in, and $a = c + s$. Moving everything in the last fact to the left gives $-c + a - s = 0$. In Unit 2-1 these became three equations:

$$\begin{cases} c + a + s = 300 \\ 6c + 10a + 8s = 2540 \\ -c + a - s = 0 \end{cases}$$

Today we will write the same problem as **one** equation, $A\bar{x} = \bar{b}$:

$$\underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 6 & 10 & 8 \\ -1 & 1 & -1 \end{bmatrix}}_A \underbrace{\begin{bmatrix} c \\ a \\ s \end{bmatrix}}_{\bar{x}} = \underbrace{\begin{bmatrix} 300 \\ 2540 \\ 0 \end{bmatrix}}_{\bar{b}}$$

Each row of A holds the coefficients of one equation: the first row counts tickets, the second row holds the ticket prices, and the third row encodes the rule about adult tickets. So A holds the **prices and rules**, $\bar{x}$ holds the unknown **counts**, and $\bar{b}$ holds the **totals**. Separating the problem into these three pieces is what lets a computer take it in as a single object. By the end of Section 5, **Matrix Times Vector**, one line of code will solve it.

It is worth reading the grid A in both directions, because the two directions tell different stories.

- **Across a row** you see one *fact* about the whole sale. Row 2, $\begin{bmatrix} 6 & 10 & 8 \end{bmatrix}$, is the price list, and the matching entry of $\bar{b}$, 2540, is the money that fact must produce.
- **Down a column** you see one *kind of ticket* across all three facts. Column 1, $\begin{bmatrix} 1 \\ 6 \\ -1 \end{bmatrix}$, says what a single child’s ticket contributes: one ticket to the count, \$6 to the takings, and -1 to the adult-rule balance.

The column reading is the vector reading of Section 1: the problem asks how many of each column it takes to reach $\bar{b}$.

Example 2.1: A vaccine clinic in three pieces. A walk-in clinic gave 120 vaccinations in one day, flu shots billed at \$30 and COVID boosters at \$50, and billed \$4400 in all. How many of

each? Let f and k be the numbers of flu shots and boosters. The two facts are

$$\begin{cases} f + k = 120 \\ 30f + 50k = 4400 \end{cases} \longleftrightarrow \underbrace{\begin{bmatrix} 1 & 1 \\ 30 & 50 \end{bmatrix}}_A \underbrace{\begin{bmatrix} f \\ k \end{bmatrix}}_{\bar{x}} = \underbrace{\begin{bmatrix} 120 \\ 4400 \end{bmatrix}}_{\bar{b}}.$$

Here A holds the *rules* (every shot counts once; each kind has its price), $\bar{x}$ the unknown *counts*, and $\bar{b}$ the *totals*, exactly the same division of labor as the matinee. Solve by substitution: $f = 120 - k$, so $30(120 - k) + 50k = 4400$, that is $3600 + 20k = 4400$, giving $k = 40$ and $f = 80$. Check against the story: $80 + 40 = 120$ shots and $30(80) + 50(40) = 2400 + 2000 = 4400$ dollars. The clinic gave 80 flu shots and 40 boosters. Notice that nothing about the *setup* would change if the clinic offered ten vaccines instead of two; A would just have more columns.

2.2 Where This Shows Up in Data Science

Almost everything you compute with in data science is a matrix.

- **A data table is a matrix.** One row per record, one column per feature. Unit 2-2 read the rows as vectors; now the whole table is one object.
- **An image is a matrix** of brightness values. Editing an image is matrix arithmetic.
- **A network is a matrix.** Contacts, friendships, web links, and gene interactions are all stored as a grid of who connects to whom.
- **Fitting a model is a matrix product.** Least-squares regression is built from $X^\top X$, where X is the data matrix and $X^\top$ is its transpose. By the end of these notes you will be able to compute that product.
- **The covariance matrix** that PCA works on at the end of this unit is a **symmetric** matrix, a type we name in Section 4.

In every case the grid means something different, but the arithmetic is the same. The table below lists what the rows, the columns, and a single entry stand for in each setting; all of these come back as examples later in these notes.

Matrix	Rows	Columns	One entry
patient data table	patients	measurements	one patient's lab value
email word counts	emails	vocabulary words	how often a word appears
grayscale image	pixel rows	pixel columns	one pixel's brightness
contact network	people	people	1 if the two met, else 0
system of equations	equations	unknowns	one coefficient

Real matrices are large. An electronic health record extract with 10,000 patients and 40 lab values is a 10,000 by 40 grid of 400,000 numbers, and a single phone photo has millions of pixels. Nobody computes with those by hand; the point of learning the arithmetic on 2 by 2 and 3 by 3 examples is to know exactly what the computer is doing, and to recognize when a result cannot be right.

Unit 2-1 already used a grid of numbers, the augmented matrix, but only as a compact way to write down a system. Today the grid becomes an object with its own arithmetic: we will add, scale, flip, and multiply grids much as we do numbers.

2.3 What We Will Build

1. Where Unit 2-2 left off: a **dependence** problem, written as $A\bar{x} = \bar{b}$ and solved.
2. A **matrix**, its **entries** a_{ij} , its **dimensions**, and the **special matrices** to know on sight.
3. **Adding, scaling, and transposing** matrices, with an image as the example.
4. **Matrix times vector**: where the rule comes from and its two readings.
5. **Matrix times matrix**: the row-by-column rule, its properties, and the one property it lacks.
6. **Powers** of a matrix, and how A^2 counts routes through a network.

Notebooks: two notebooks go with these notes.

- U2-3_Matrices-1_ArithmeticAndSpecial, for Sections 3 and 4.
- U2-3_Matrices-2_MultiplicationAndPowers, for Sections 5 to 7.

2.4 Extra Practice

Exercise 2.1

A clinic's supply order bought gauze packs at \$4 each, tape rolls at \$3, and saline bottles at \$5: 50 items in all, for \$190, with twice as many gauze packs as tape rolls.

1. Write the three facts as equations in g , t , and s .
2. Name the pieces A , $\bar{x}$, and $\bar{b}$ of $A\bar{x} = \bar{b}$, and say in words what each row of A stands for.
3. Solve the system by substitution and check the answer against the story.

Solution to Exercise 2.1

1. The count, the cost, and the ratio: $g + t + s = 50$, $4g + 3t + 5s = 190$, and $g = 2t$. Move everything in the last one to the left so that every unknown sits on the left and a number on the right: $g - 2t = 0$.
2. One row per equation, one column per unknown, in the order g, t, s in every row (the ratio equation has no s , so its third entry is 0):

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 4 & 3 & 5 \\ 1 & -2 & 0 \end{bmatrix}, \quad \bar{x} = \begin{bmatrix} g \\ t \\ s \end{bmatrix}, \quad \bar{b} = \begin{bmatrix} 50 \\ 190 \\ 0 \end{bmatrix}.$$

Row 1 counts items, row 2 is the price list, and row 3 encodes the “twice as many” rule. Column 1 is what one gauze pack contributes to each fact.

3. Substitute $g = 2t$. The count becomes $3t + s = 50$, so $s = 50 - 3t$. The cost becomes $8t + 3t + 5(50 - 3t) = 190$, that is $250 - 4t = 190$, so $t = 15$. Then $g = 30$ and $s = 50 - 45 = 5$.

Check: $30 + 15 + 5 = 50$ items; $4(30) + 3(15) + 5(5) = 120 + 45 + 25 = 190$ dollars; and $30 = 2(15)$. The order was 30 gauze packs, 15 tape rolls, and 5 bottles of saline.

3 Matrices and Their Entries

3.1 Definition and Notation

Think of a spreadsheet with every label stripped off: no column headers, no row names, just the numbers left in their cells. That is a matrix. The numbers keep their *positions*, and the positions carry the meaning: which row a number sits in and which column it sits in are the only clues to what it stands for.

Definition: Matrix

An $m \times n$ **matrix** is a rectangular grid of numbers with m **rows** and n **columns**. The entry in row i , column j is written a_{ij} , and the whole matrix is written $A = [a_{ij}]$.

Rows run across the page and columns run down it. The usual habit is to name a matrix with a capital letter and its entries with the matching lowercase letter: the entries of A are a_{ij} , the entries of M are m_{ij} , and so on. Here is a concrete matrix next to the general pattern of the same shape:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \qquad A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

The left one is a 2×3 matrix: two rows, three columns. Lining it up with the pattern, $a_{12} = 2$ (row 1, column 2) and $a_{23} = 6$ (row 2, column 3). The first subscript is always the **row** and the second the **column**. Rows first, always.

Two conventions are worth fixing now, because everything later depends on them.

- **Dimensions are stated rows $\times$ columns.** A 3×1 matrix is tall (three rows, one column), and a 1×3 matrix is wide (one row, three columns). A matrix with $m = n$, as many rows as columns, is called **square**.
- **A vector is a matrix.** The column vector $\bar{v} \in \mathbb{R}^n$ from Unit 2-2 is simply an $n \times 1$ matrix. Its transpose $\bar{v}^\top$, the same entries tipped on their side into a row as in the dot product of Unit 2-2, is a $1 \times n$ matrix. (Section 4 defines the transpose for every matrix.) So nothing new has to be learned for vectors: every rule in these notes applies to them.

In code, `A.shape` is `(2, 3)`, rows first, and `A[0, 1]` is a_{12} : Python counts from 0, mathematics from 1. So `A[0, 1]` means “the first row, the second column.” This off-by-one is the most common bug in a first linear algebra notebook.

Example 3.1: Shapes at a glance. State the dimensions of each matrix.

$$\begin{bmatrix} 4 & -1 \\ 0 & 2 \\ 7 & 3 \end{bmatrix} \qquad \begin{bmatrix} 5 & 0 & -2 & 1 \end{bmatrix} \qquad \begin{bmatrix} 3 \\ 8 \end{bmatrix} \qquad \begin{bmatrix} 7 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & 2 \\ 4 & -3 & 5 \\ 6 & 1 & 0 \end{bmatrix}$$

Count rows first (how many lines of numbers run across), then columns (how many run down).

- The first is 3×2 : three rows, two columns. It is taller than it is wide.
- The second is 1×4 : a single row, a **row vector** with four entries.
- The third is 2×1 : a single column, which is exactly a vector in $\mathbb{R}^2$ as Unit 2-2 wrote it.
- The fourth is 1×1 : one row, one column, one number. A 1×1 matrix behaves just like the ordinary number inside it, and we will treat the two as the same thing.
- The last is 3×3 , and **square**.

The shape is the first thing to check about any matrix, before looking at a single entry, because (as the rest of these notes show) the shape decides which operations are even allowed.

Not every grid-shaped arrangement of things is a matrix. Two **non-examples**:

- **A ragged list**, such as a first row with three numbers and a second row with only two. A matrix must be a full rectangle: every row the same length, every column the same height. A spreadsheet with a missing value has a hole in it, and it must be filled in (or the row dropped) before it can be treated as a matrix; this is one reason data cleaning comes first in every analysis.
- **A table containing words**, such as a column of sex recorded as “F” and “M”. A matrix holds numbers only. Such a column has to be *coded* as numbers (for instance 1 and 0) before the table becomes a matrix.

Example 3.2: Reading entries. Let

$$B = \begin{bmatrix} 2 & -1 & 0 & 7 \\ 5 & 3 & 8 & -2 \\ 0 & 4 & 1 & 6 \end{bmatrix}.$$

B has three rows and four columns, so it is 3×4 . To read an entry b_{ij} , go down to row i first, then across to column j .

- b_{23} : row 2 is $\begin{bmatrix} 5 & 3 & 8 & -2 \end{bmatrix}$, and its third entry is $b_{23} = 8$.
- b_{32} : row 3 is $\begin{bmatrix} 0 & 4 & 1 & 6 \end{bmatrix}$, and its second entry is $b_{32} = 4$. Swapping the subscripts changed the answer from 8 to 4: order matters.
- $b_{14} = 7$, the last entry of the first row, and $b_{31} = 0$, the first entry of the last row.
- b_{41} **does not exist**: B has no fourth row. (It does have a fourth column, which is what you would reach if you read the subscripts in the wrong order.)

Whole rows and columns can be pulled out too. Row 2 of B is the 1×4 row $\begin{bmatrix} 5 & 3 & 8 & -2 \end{bmatrix}$, and column 3 is the 3×1 column $\begin{bmatrix} 0 \\ 8 \\ 1 \end{bmatrix}$. In code, $B[1, 2]$ is $b_{23} = 8$, $B[1, :]$ is row 2, and $B[:, 2]$ is column 3.

Example 3.3: Building a matrix from a rule. Sometimes a matrix is described by a formula for its entries rather than by a list. Write out the 2×3 matrix with $a_{ij} = i + j$, and the 3×3 matrix with $c_{ij} = i - j$.

For A , the row number i runs from 1 to 2 and the column number j from 1 to 3. Fill in one row at a time: row 1 is $1 + 1, 1 + 2, 1 + 3$ and row 2 is $2 + 1, 2 + 2, 2 + 3$. For C , each entry is “row number minus column number.”

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}, \quad C = \begin{bmatrix} 1-1 & 1-2 & 1-3 \\ 2-1 & 2-2 & 2-3 \\ 3-1 & 3-2 & 3-3 \end{bmatrix} = \begin{bmatrix} 0 & -1 & -2 \\ 1 & 0 & -1 \\ 2 & 1 & 0 \end{bmatrix}.$$

Two patterns are visible. In C , every entry on the line from the top left to the bottom right is 0, because there $i = j$. And the entries of C on either side of that line are negatives of each other ($c_{21} = 1$ and $c_{12} = -1$), because swapping i and j flips the sign of $i - j$. Building a matrix from a rule is good practice precisely because it forces you to keep i as the row and j as the column.

Watch out: mathematics counts from 1, Python from 0

Every entry, row, and column is shifted by one between the page and the code:

On paper	In numpy
the entry a_{11}	<code>A[0, 0]</code>
the entry a_{23}	<code>A[1, 2]</code>
the entry a_{ij}	<code>A[i-1, j-1]</code>
row i	<code>A[i-1, :]</code>
column j	<code>A[:, j-1]</code>
an $m \times n$ matrix	<code>A.shape == (m, n)</code>

The order (row, then column) is the same in both places; only the starting number differs. When a notebook result looks shifted by one row or one column, this is the first thing to check.

Notebook U2-3_Matrices-1_ArithmeticAndSpecial, §1, entries and dimensions

3.2 A Data Table Is a Matrix

Here are three patients with three measurements each: age in years, blood pressure (BP), and A1c. The first two patients are Unit 2-2’s patient vectors $\bar{p}$ and $\bar{q}$. Keep only the numbers and you have a matrix:

	Age	BP	A1c		
Patient 1	54	130	6.1	$\xrightarrow{\text{numbers only}}$	$X = \begin{bmatrix} 54 & 130 & 6.1 \\ 58 & 138 & 6.4 \\ 41 & 118 & 5.6 \end{bmatrix}$
Patient 2	58	138	6.4		
Patient 3	41	118	5.6		

The same grid can be read two ways.

- Each **row** is one patient: row 1 is patient 1’s measurements.
- Each **column** is one feature across everyone: the age column holds three numbers, one per patient, so it is a vector in $\mathbb{R}^3$.
- The entry x_{ij} is feature j of patient i . For example, $x_{23} = 6.4$ is patient 2’s A1c.

Both readings matter, because they answer different questions. To find **similar patients**, compare rows. To find **redundant features**, compare columns. One more point about orientation: one sample per row is how data is **stored**, but when we pull a single patient out of the table to work with as a vector, we still write it as a column, as in Unit 2-2.

Notice what the matrix has thrown away: the labels. X does not know that its second column is blood pressure. That is the price of the compact form, and it is why a notebook keeps the column names alongside the numbers (a **pandas** data frame is exactly a matrix plus its labels). When you do arithmetic on X , it is your job to remember what each position means.

Example 3.4: Emails as a word-count matrix. Unit 2-2 turned three emails into count vectors over the vocabulary *free, win, cash, meeting*:

$$\bar{d}_1 = \begin{bmatrix} 2 \\ 1 \\ 2 \\ 0 \end{bmatrix}, \quad \bar{d}_2 = \begin{bmatrix} 4 \\ 2 \\ 4 \\ 0 \end{bmatrix}, \quad \bar{d}_3 = \begin{bmatrix} 2 \\ 0 \\ 1 \\ 2 \end{bmatrix}.$$

Stored as a data table, one email per **row** and one word per **column**, they become a 3×4 matrix:

	<i>free</i>	<i>win</i>	<i>cash</i>	<i>meeting</i>
Email 1	2	1	2	0
Email 2	4	2	4	0
Email 3	2	0	1	2

 $\xrightarrow{\text{numbers only}}$

$$E = \begin{bmatrix} 2 & 1 & 2 & 0 \\ 4 & 2 & 4 & 0 \\ 2 & 0 & 1 & 2 \end{bmatrix}$$

Now read it both ways.

- **An entry:** $e_{34} = 2$ says email 3 contains *meeting* twice; $e_{22} = 2$ says email 2 contains *win* twice.
- **Rows (compare emails):** row 2, $\begin{bmatrix} 4 & 2 & 4 & 0 \end{bmatrix}$, is exactly twice row 1, $\begin{bmatrix} 2 & 1 & 2 & 0 \end{bmatrix}$. Email 2 has the same word profile as email 1, only longer, which is why Unit 2-2 found their cosine similarity to be 1.
- **Columns (compare words):** the *free* column $\begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix}$ and the *cash* column $\begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix}$ are nearly the same. Across these emails the two words travel together, so as features they are nearly redundant; a spam filter learns little from the second once it has the first.

One more observation: Unit 2-2 drew these emails as *columns* (one column per email), while the data table has them as *rows*. The two grids hold the same twelve numbers, flipped over. Section 4 names that flip: it is the transpose.

Example 3.5: A lab panel. A clinic records fasting glucose, LDL cholesterol, and creatinine (all in mg/dL) for four patients:

$$L = \begin{bmatrix} 92 & 110 & 0.9 \\ 126 & 145 & 1.1 \\ 101 & 98 & 0.8 \\ 140 & 160 & 1.4 \end{bmatrix} \quad \begin{array}{l} \leftarrow \text{patient 1} \\ \leftarrow \text{patient 2} \\ \leftarrow \text{patient 3} \\ \leftarrow \text{patient 4} \end{array}$$

with columns glucose, LDL, creatinine, in that order.

- **Shape:** four patients and three labs, so L is 4×3 .
- **Entries:** $\ell_{42} = 160$ is patient 4's LDL, and $\ell_{23} = 1.1$ is patient 2's creatinine. (In code: `L[3, 1]` and `L[1, 2]`.)
- **A column:** column 1, $\begin{bmatrix} 92 & 126 & 101 & 140 \end{bmatrix}^\top$ (the four glucose values stacked as a column), is glucose across the clinic. Its average is a meaningful summary: the clinic's mean fasting glucose.
- **A row, as a vector:** patient 3 is row 3 of the table, but pulled out on its own it is written as the column $\begin{bmatrix} 101 \\ 98 \\ 0.8 \end{bmatrix} \in \mathbb{R}^3$.

The shape and the labels together tell you what any single number means: ℓ_{ij} is “lab j of patient i .”

Watch out: not every sum of entries means something

Averaging *down a column* of L combines like with like: glucose values from different patients. Adding *across a row* does not: glucose plus LDL plus creatinine is a number with no clinical meaning, even though all three happen to be in mg/dL. Unit 2-2 made the same point about the length of a patient vector. Before combining entries of a data matrix, ask whether they measure the same thing.

3.3 Equality, and Special Matrices to Know on Sight

Two matrices are **equal** when they have the same dimensions **and** every entry agrees. The shape check comes first: a 2×3 matrix is never equal to a 3×2 one, whatever the entries, even if the two grids contain the same six numbers.

Equality of matrices is a very strong statement. An equation between two 2×2 matrices is really *four* equations between numbers, one per position, and that makes it a handy way to pack a system of equations.

Example 3.6: Using equality to find unknown entries. Find the unknowns, or explain why there are none.

$$1. \begin{bmatrix} x & 2 \\ 3 & y \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 3 & -1 \end{bmatrix}$$

Match positions: the top-left entries give $x = 5$, the bottom-right give $y = -1$, and the other two positions ($2 = 2$ and $3 = 3$) already agree. So $x = 5$ and $y = -1$.

$$2. \begin{bmatrix} x+y & 3 \\ x-y & 0 \end{bmatrix} = \begin{bmatrix} 7 & 3 \\ 1 & 0 \end{bmatrix}$$

Matching the first column gives a small system, $x + y = 7$ and $x - y = 1$. Adding the two equations gives $2x = 8$, so $x = 4$, and then $y = 7 - 4 = 3$. One matrix equation has become a Unit 2-1 system.

$$3. \begin{bmatrix} x & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$$

The top-left position wants $x = 1$, but the bottom-right position says $4 = 5$, which is false

no matter what x is. **No solution:** one disagreeing position is enough to make two matrices unequal.

$$4. \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \stackrel{?}{=} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Never, even though the numbers match in order: one is 1×3 and the other 3×1 . Shape comes before entries.

In words: equality demands agreement at *every* position, and a shape mismatch rules it out before any number is looked at.

A few matrices come up so often that they have names. To describe them we need one more word. The **diagonal** (or main diagonal) of a matrix is the entries $a_{11}, a_{22}, a_{33}, \dots$, running from the top left toward the bottom right.

Definition: Three named types

Name	Rule	Example
zero , 0	every entry is 0	$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$
identity , I	1s on the diagonal, 0 elsewhere	$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
diagonal	nonzero only on the diagonal	$\begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}$

“Nonzero only on the diagonal” means that every entry *off* the diagonal is 0. By that rule the identity is a special diagonal matrix, one whose diagonal entries are all 1. The identity is written I_n when its size needs saying: I_2 is the one in the table, and I_3 is the 3×3 version.

In words: the **zero matrix** is a grid with nothing in it; the **identity** is the grid with a line of 1s down its main diagonal and nothing else; a **diagonal matrix** allows any numbers on that line but nothing off it. The identity and diagonal matrices are always square in this course (a diagonal line from corner to corner needs as many rows as columns), but a zero matrix comes in every shape.

Example 3.7: The named types in other sizes.

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad 0_{2 \times 3} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 7 \end{bmatrix}.$$

I_3 is the 3×3 identity. The zero matrix $0_{2 \times 3}$ is not square, and that is fine: there is a zero matrix of every shape, and the symbol 0 means whichever one fits the context. D is **diagonal** even though one of its diagonal entries is 0: the rule restricts the entries *off* the diagonal (they must be 0) and leaves the diagonal entries free, zero included.

It helps to sort a few matrices, including some that almost qualify.

Matrix	zero?	identity?	diagonal?	Why
$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$	yes	no	yes	every off-diagonal entry is 0, so it passes the diagonal test too
$\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$	no	no	yes	3s, not 1s, on the diagonal
$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	no	no	no	its 1s sit on the <i>other</i> diagonal, which does not count
$\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$	no	no	no	the 2 is off the diagonal
$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$	no	no	no	not square, so not an identity, even though I_2 sits inside it

Watch out: “diagonal” means the main diagonal

Only the line from the **top left** to the **bottom right** is the diagonal. A matrix like $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, with its entries on the line from top right to bottom left, is **not** diagonal, however neat it looks. And the test for “diagonal” is about what sits *off* the diagonal, not on it.

For now these are shapes to recognize and nothing more. They earn their names once multiplication arrives: Section 5 shows what the identity and a diagonal matrix actually *do*.

Notebook U2-3_Matrices-1_ArithmeticAndSpecial, §4, the special types displayed

3.4 Exercise: Read the Matrix

Exercise 3.1

Let

$$M = \begin{bmatrix} 3 & 0 & -1 \\ 2 & 7 & 4 \end{bmatrix}, \quad S = \begin{bmatrix} 2 & 5 \\ 5 & -1 \end{bmatrix}, \quad D = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}.$$

1. State the dimensions of M , and read off m_{13} and m_{21} .
2. Which of S and D is diagonal? Is either the identity?
3. Can M ever equal S ? Why or why not?
4. A patient record with 12 lab values is stored for 250 patients. What are the dimensions of the data matrix, and what does the entry in row 17, column 3 mean?

Solution to Exercise 3.1

1. M has two rows and three columns, so it is 2×3 . For m_{13} , go to row 1, which is 3, 0, -1 , and take its third entry: $m_{13} = -1$. For m_{21} , go to row 2, which is 2, 7, 4, and take its first entry: $m_{21} = 2$. Not the other way round: rows first. (Reading m_{13} as “column 1, row 3” would not even make sense, since M has no third row.)
2. A diagonal matrix must have 0 in every position off the diagonal. S has 5s off the diagonal, so it is **not** diagonal. D has 0s off the diagonal, so it **is** diagonal. Neither is the identity: the identity must also have 1s on the diagonal, and D has 4s there. In fact

$D = 4I$, the identity with every entry multiplied by 4 (an operation Section 4 calls scalar multiplication). We will see in Section 5 that such a matrix stretches everything by 4.

3. **Never.** M is 2×3 and S is 2×2 . Equality requires the same shape before a single entry is compared, so no choice of numbers could make them equal.
4. One row per patient and one column per lab value gives 250 rows and 12 columns, so the data matrix is 250×12 (rows first, not 12×250). The entry in row 17, column 3 is the **third lab value of the seventeenth patient**. Reading more widely, the whole of row 17 is that patient's feature vector, 12 numbers long, and the whole of column 3 is that one lab value across all 250 patients.

3.5 Extra Practice

Exercise 3.2

Write out the 3×3 matrix A with entries $a_{ij} = 2i - j$. Then read off a_{31} and a_{13} , give the numpy index for a_{31} , and say whether A is diagonal.

Solution to Exercise 3.2

Fill in one row at a time, holding i fixed while j runs from 1 to 3. Row 1 is $2 - 1, 2 - 2, 2 - 3$; row 2 is $4 - 1, 4 - 2, 4 - 3$; row 3 is $6 - 1, 6 - 2, 6 - 3$:

$$A = \begin{bmatrix} 1 & 0 & -1 \\ 3 & 2 & 1 \\ 5 & 4 & 3 \end{bmatrix}.$$

Then $a_{31} = 2(3) - 1 = 5$ (row 3, column 1) and $a_{13} = 2(1) - 3 = -1$ (row 1, column 3). Swapping the subscripts gave a different number, as it usually does. In `numpy`, a_{31} is `A[2, 0]`. A is **not** diagonal: almost every off-diagonal entry is nonzero (only $a_{12} = 0$).

Exercise 3.3

Find x , y , and z so that

$$\begin{bmatrix} x & 2y \\ z & 4 \end{bmatrix} = \begin{bmatrix} 3 & 8 \\ x + y & 4 \end{bmatrix}.$$

Solution to Exercise 3.3

Both matrices are 2×2 , so equality is possible; match the four positions. Top left: $x = 3$. Top right: $2y = 8$, so $y = 4$. Bottom left: $z = x + y = 3 + 4 = 7$. Bottom right: $4 = 4$, already true. So $x = 3$, $y = 4$, $z = 7$. Notice that the equations had to be solved in a sensible order: the bottom-left position depended on the answers from the top row.

Exercise 3.4

A health survey asks 25 questions of 1200 respondents, and the answers are stored with one respondent per row.

1. What are the dimensions of the data matrix X ?
2. What do $x_{5,12}$, row 5, and column 12 each mean? Which **numpy** expression gives $x_{5,12}$?
3. A colleague stores the same answers with one *question* per row. Is their matrix equal to X ?

Solution to Exercise 3.4

1. 1200 rows (respondents) and 25 columns (questions): X is 1200×25 .
2. $x_{5,12}$ is respondent 5's answer to question 12. Row 5 is all 25 of respondent 5's answers, that person's feature vector. Column 12 is every respondent's answer to question 12, one feature across the whole sample. In **numpy** the entry is `X[4, 11]`.
3. **No.** Their matrix is 25×1200 , a different shape, so it cannot equal X even though it holds the same 30,000 numbers. It is X flipped over its diagonal, which Section 4 calls the transpose.

Exercise 3.5

Classify each matrix as zero, identity, diagonal, or none of these (more than one label can apply):

$$\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 \\ 0 & 5 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Solution to Exercise 3.5

1. **Diagonal**, since both off-diagonal entries are 0. Not the identity (3s, not 1s); it is $3I_2$.
2. **Zero** (2×3). It is not square, so the words identity and diagonal do not apply.
3. **None**: the 1 in the top-right position is off the diagonal. It has 1s on the diagonal, but that is not enough to be the identity; the off-diagonal entries must be 0 too.
4. **Diagonal**. The 0 in the top-left sits *on* the diagonal, which is allowed; only off-diagonal entries must be 0.
5. **Diagonal**, and not the identity: the -1 is on the diagonal, where the identity needs a 1.

4 Matrix Arithmetic

4.1 Addition and Scalar Multiplication

Matrices add and scale entry by entry, exactly as vectors did in Unit 2-2. Recall that a **scalar** is a single ordinary number. Picture laying one grid on top of another of the same size: each position lines up with exactly one position in the other grid, and the two numbers there are added. Scaling is even simpler: every position is multiplied by the same number.

Definition: Addition and scalar multiplication

Matrices of the **same shape** add entry by entry, and a scalar multiplies every entry:

$$(A + B)_{ij} = a_{ij} + b_{ij}, \quad (cA)_{ij} = c a_{ij}.$$

Subtraction is $A - B = A + (-1)B$.

Read the first formula as: “the entry of $A + B$ in row i , column j is the sum of the entries of A and B in that same position.” The second says that cA is A with every entry multiplied by c . The vector rules of Unit 2-2 are the special case of a single column: adding two $n \times 1$ matrices is adding two vectors.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}, \quad 2 \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$$

In the sum, for example, the bottom-left entry is $3 + 7 = 10$; in the scalar multiple, every entry has simply doubled.

Example 4.1: A sum with negatives, and not square.

$$\begin{bmatrix} 1 & -2 & 0 \\ 4 & 3 & -1 \end{bmatrix} + \begin{bmatrix} 3 & 5 & -2 \\ 0 & -3 & 6 \end{bmatrix} = \begin{bmatrix} 1+3 & -2+5 & 0+(-2) \\ 4+0 & 3+(-3) & -1+6 \end{bmatrix} = \begin{bmatrix} 4 & 3 & -2 \\ 4 & 0 & 5 \end{bmatrix}.$$

Both matrices are 2×3 , so the sum exists and is 2×3 as well. Nothing about addition needs a square matrix; it only needs the two shapes to match. Writing the middle step, with every pair laid out, is the surest way to avoid sign slips like $3 + (-3)$.

Example 4.2: Clinic visits over two months. A health system has two clinics, North and South, and tracks three kinds of visit: primary care, dental, and lab. Each month’s counts form a 2×3 matrix, rows for clinics and columns for visit types:

$$J = \begin{bmatrix} 120 & 45 & 80 \\ 95 & 30 & 60 \end{bmatrix} \text{ (January)}, \quad F = \begin{bmatrix} 110 & 50 & 90 \\ 100 & 35 & 55 \end{bmatrix} \text{ (February)}.$$

Three natural questions are three pieces of matrix arithmetic.

- **Total for the two months:** add position by position.

$$J + F = \begin{bmatrix} 120 + 110 & 45 + 50 & 80 + 90 \\ 95 + 100 & 30 + 35 & 60 + 55 \end{bmatrix} = \begin{bmatrix} 230 & 95 & 170 \\ 195 & 65 & 115 \end{bmatrix}.$$

- **Change from January to February:** subtract.

$$F - J = \begin{bmatrix} 110 - 120 & 50 - 45 & 90 - 80 \\ 100 - 95 & 35 - 30 & 55 - 60 \end{bmatrix} = \begin{bmatrix} -10 & 5 & 10 \\ 5 & 5 & -5 \end{bmatrix}.$$

- **Monthly average:** add, then scale by $\frac{1}{2}$.

$$\frac{1}{2}(J + F) = \begin{bmatrix} 115 & 47.5 & 85 \\ 97.5 & 32.5 & 57.5 \end{bmatrix}.$$

Each answer keeps the same layout, so it is read the same way: in $F - J$, the -10 in row 1, column 1 says North saw 10 fewer primary-care visits in February, and the -5 in row 2, column 3 says South's lab visits fell by 5. Matrix arithmetic did every clinic and every visit type in one step. The average is a *linear combination* of matrices, $\frac{1}{2}J + \frac{1}{2}F$, exactly as Unit 2-2 combined vectors.

Example 4.3: Converting units with a scalar. Four fasting glucose readings in mg/dL, two patients (rows) on two days (columns), are

$$G = \begin{bmatrix} 90 & 108 \\ 126 & 144 \end{bmatrix}.$$

To convert glucose from mg/dL to mmol/L, divide by 18 (the standard conversion factor, rounded). Dividing by 18 is multiplying by the scalar $\frac{1}{18}$:

$$\frac{1}{18}G = \begin{bmatrix} 90/18 & 108/18 \\ 126/18 & 144/18 \end{bmatrix} = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} \text{ mmol/L}.$$

Every reading was converted at once. This worked because every entry of G is a glucose value in the same unit, so one factor suits them all.

Example 4.4: A linear combination of matrices. Let $A = \begin{bmatrix} 2 & -1 \\ 0 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 4 \\ -2 & 5 \end{bmatrix}$. Compute $3A - 2B$. Scale first, then subtract position by position:

$$3A = \begin{bmatrix} 6 & -3 \\ 0 & 9 \end{bmatrix}, \quad 2B = \begin{bmatrix} 2 & 8 \\ -4 & 10 \end{bmatrix}, \quad 3A - 2B = \begin{bmatrix} 6-2 & -3-8 \\ 0-(-4) & 9-10 \end{bmatrix} = \begin{bmatrix} 4 & -11 \\ 4 & -1 \end{bmatrix}.$$

The bottom-left entry is where students most often slip: subtracting a negative, $0 - (-4)$, gives $+4$. Doing the scaling in its own step, as here, keeps each operation simple.

Because each entry is an ordinary number, and these operations work on one position at a time, the familiar rules of arithmetic carry over unchanged.

Rules that carry over from numbers

- $A + B = B + A$ (order does not matter)
- $(A + B) + C = A + (B + C)$ (grouping does not matter)
- $c(A + B) = cA + cB$, and $A + 0 = A$

Each rule holds for the same reason: it holds for numbers, and matrix addition and scaling are just many number additions and multiplications done side by side. For instance, with A and B from the last example, $A + B$ and $B + A$ are both $\begin{bmatrix} 3 & 3 \\ -2 & 8 \end{bmatrix}$, since at each position $a_{ij} + b_{ij} = b_{ij} + a_{ij}$. In the rule $A + 0 = A$, the 0 is the zero matrix of the same shape as A .

These rules let us solve simple equations whose unknown is a whole matrix, by the same moves as in ordinary algebra.

Example 4.5: Solving for an unknown matrix. Find the matrix X with $2X + A = B$, where $A = \begin{bmatrix} 1 & 3 \\ 5 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 7 & 3 \\ 1 & 9 \end{bmatrix}$. Treat it like $2x + a = b$ with numbers: subtract A from both sides, then scale both sides by $\frac{1}{2}$.

$$2X = B - A = \begin{bmatrix} 7-1 & 3-3 \\ 1-5 & 9-(-1) \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ -4 & 10 \end{bmatrix}, \quad X = \frac{1}{2} \begin{bmatrix} 6 & 0 \\ -4 & 10 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ -2 & 5 \end{bmatrix}.$$

Check by substituting back: $2X + A = \begin{bmatrix} 6 & 0 \\ -4 & 10 \end{bmatrix} + \begin{bmatrix} 1 & 3 \\ 5 & -1 \end{bmatrix} = \begin{bmatrix} 7 & 3 \\ 1 & 9 \end{bmatrix} = B$. ✓ A matrix equation of this kind is really four ordinary equations, one per position, solved all at once.

Watch out: the shapes must match

A 2×2 matrix cannot be added to a 2×3 one. There is no entry to pair the extra column with, so the sum is **undefined**. It is not “padded with zeros” to make the shapes fit; it simply does not exist. In **numpy**, trying it raises a broadcast error.

Watch out: one scalar scales every entry

Scalar multiplication cannot convert just *one column* of a data table. In the patient table X of Section 3, multiplying by any single number changes the ages, the blood pressures, and the A1c values alike. Converting one feature while leaving the others alone needs a different tool, a matrix product that Example 6.8 in Section 6 builds.

Notebook U2-3_Matrices-1_ArithmeticAndSpecial, §2, sums and scalar multiples

4.2 The Transpose

The transpose is the first operation with no counterpart among ordinary numbers. It does not change any value; it only moves the values around.

Definition: Transpose

The **transpose** $A^\top$ swaps rows and columns: $(A^\top)_{ij} = a_{ji}$. The grid flips over its main diagonal, so an $m \times n$ matrix becomes $n \times m$.

In practice: row 1 of A becomes column 1 of $A^\top$, row 2 becomes column 2, and so on. The formula says the same thing entry by entry: whatever sits in row j , column i of A moves to row i , column j of $A^\top$. Entries on the diagonal, where $i = j$, stay where they are.

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}^\top = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} \qquad \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}^\top = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

On the left, the rows 1, 2, 3 and 4, 5, 6 have become the columns, and the 2×3 matrix has become 3×2 . On the right, the diagonal entries 1 and 4 stay put while 2 and 3 trade places across the diagonal.

Example 4.6: Three transposes, three shapes.

$$\begin{bmatrix} 2 & -1 & 5 & 0 \end{bmatrix}^\top = \begin{bmatrix} 2 \\ -1 \\ 5 \\ 0 \end{bmatrix}, \qquad \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}^\top = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}, \qquad \begin{bmatrix} 4 & -1 \\ 0 & 2 \\ 7 & 3 \end{bmatrix}^\top = \begin{bmatrix} 4 & 0 & 7 \\ -1 & 2 & 3 \end{bmatrix}.$$

- The 1×4 row becomes a 4×1 column: the same four numbers, in the same order, now stacked. This is the “tipping over” of Unit 2-2, run in the other direction.

- The square 3×3 matrix stays 3×3 . The diagonal 1, 5, 9 does not move, and every other entry swaps with its mirror image: the 2 in position (1, 2) and the 4 in position (2, 1) trade places, and so do 3 with 7 and 6 with 8.
- The 3×2 matrix becomes 2×3 . Track one entry to see the formula at work: the 7 sits in row 3, column 1 of the original, so it lands in row 1, column 3 of the transpose.

In all three cases, reading the original *down* its columns gives exactly what the transpose shows *across* its rows.

Example 4.7: Two ways to store the emails. The word-count matrix E of Section 3 stores one email per row. Its transpose stores one email per column:

$$E = \begin{bmatrix} 2 & 1 & 2 & 0 \\ 4 & 2 & 4 & 0 \\ 2 & 0 & 1 & 2 \end{bmatrix}, \quad E^\top = \begin{bmatrix} 2 & 4 & 2 \\ 1 & 2 & 0 \\ 2 & 4 & 1 \\ 0 & 0 & 2 \end{bmatrix}.$$

The columns of $E^\top$ are exactly the count vectors $\bar{d}_1, \bar{d}_2, \bar{d}_3$ of Unit 2-2, and $E^\top$ is the table Unit 2-2 drew. Nothing about the data changed, only its orientation: E is 3×4 (emails by words) and $E^\top$ is 4×3 (words by emails). Different software, and different formulas, expect different orientations, and the transpose is how you convert between them.

Rules for the transpose

- $(A^\top)^\top = A$: flipping twice returns the original.
- $(A + B)^\top = A^\top + B^\top$
- $(cA)^\top = cA^\top$
- A column $\bar{v}$ tipped over is the row $\bar{v}^\top$, as in Unit 2-2.

Example 4.8: Checking a rule on numbers. Take $A = \begin{bmatrix} 2 & -1 \\ 0 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 4 \\ -2 & 5 \end{bmatrix}$ again. Add first, then transpose:

$$A + B = \begin{bmatrix} 3 & 3 \\ -2 & 8 \end{bmatrix}, \quad (A + B)^\top = \begin{bmatrix} 3 & -2 \\ 3 & 8 \end{bmatrix}.$$

Or transpose first, then add:

$$A^\top + B^\top = \begin{bmatrix} 2 & 0 \\ -1 & 3 \end{bmatrix} + \begin{bmatrix} 1 & -2 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ 3 & 8 \end{bmatrix}.$$

The same matrix either way, as the rule promises. The reason is that the transpose only moves entries, and addition works position by position, so it does not matter whether the entries are moved before or after they are added. The rule $(cA)^\top = cA^\top$ holds for the same reason.

Watch out: a transpose is a flip, not a rotation

It is tempting to picture $A^\top$ as A turned a quarter turn. It is not. Turning $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ a quarter turn clockwise gives $\begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix}$, but the transpose is $\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$. A transpose **flips** the grid over its main diagonal, like turning a transparency over along that line, so the diagonal entries never move.

A check that always works: row 1 of A must reappear, in order, as column 1 of $A^\top$.

Some matrices are unchanged by the flip, and they get a name.

Definition: Symmetric matrix

A matrix A is **symmetric** when $A^\top = A$, that is, when $a_{ij} = a_{ji}$ for every i and j .

A symmetric matrix is a mirror image of itself across the diagonal. In $\begin{bmatrix} 1 & 4 \\ 4 & 3 \end{bmatrix}$, the 4 above the diagonal is matched by a 4 below it, so flipping changes nothing. Only a square matrix can be symmetric, since transposing an $m \times n$ matrix with $m \neq n$ changes its shape, and matrices of different shapes are never equal.

The identity and every diagonal matrix are symmetric: every entry off the diagonal is 0, so each mirror pair matches ($0 = 0$), and the diagonal itself never moves.

Example 4.9: Checking for symmetry. Is each matrix symmetric? Check each mirror pair across the diagonal.

$$S = \begin{bmatrix} 4 & 1 & -2 \\ 1 & 0 & 5 \\ -2 & 5 & 3 \end{bmatrix}, \quad N = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}, \quad R = \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 0 & 0 \end{bmatrix}.$$

- S : the pairs are $s_{12} = s_{21} = 1$, $s_{13} = s_{31} = -2$, and $s_{23} = s_{32} = 5$. All three match, so S **is symmetric**. A 3×3 matrix has exactly three mirror pairs to check; the diagonal 4, 0, 3 needs no check.
- N : $n_{12} = 2$ but $n_{21} = 3$. One mismatched pair is enough: N is **not symmetric**. (Equal diagonal entries do not help; symmetry is about the entries *off* the diagonal.)
- R : not square, so **not symmetric**, and no checking of entries is needed. Its transpose is 2×3 , a different shape.

Example 4.10: Symmetric tables in health data. Symmetric matrices appear whenever a table records a relationship that runs both ways.

- **Distances.** The driving distances, in miles, between three clinics:

$$\begin{bmatrix} 0 & 12 & 7 \\ 12 & 0 & 9 \\ 7 & 9 & 0 \end{bmatrix}.$$

The distance from clinic 1 to clinic 2 is the distance from clinic 2 to clinic 1, so the table is symmetric, and each clinic is 0 miles from itself.

- **Similarities.** Put the cosine similarities of the three emails from Unit 2-2 in a table, with entry (i, j) the cosine between emails i and j :

$$\begin{bmatrix} 1 & 1 & \frac{2}{3} \\ 1 & 1 & \frac{2}{3} \\ \frac{2}{3} & \frac{2}{3} & 1 \end{bmatrix}.$$

Every vector has cosine 1 with itself, so the diagonal is all 1s, and $\cos \theta$ between emails i and j is the same as between j and i (the dot product does not care about order), so the table is symmetric. The patient similarity heatmap of the Unit 2-2 notebook is the same kind of table, just larger.

- **A non-example: referrals.** Let entry (i, j) count the patients clinic i referred to clinic j last month:

$$\begin{bmatrix} 0 & 5 & 2 \\ 1 & 0 & 4 \\ 0 & 3 & 0 \end{bmatrix}.$$

Clinic 1 sent 5 patients to clinic 2, but clinic 2 sent only 1 back. A relationship with a direction gives a table that is generally **not** symmetric.

The pattern to remember: “from i to j ” equals “from j to i ” exactly when the table is symmetric.

Example 4.11: Completing a symmetric matrix. Find a , b , and c so that the matrix on the left is symmetric.

$$\begin{bmatrix} 1 & a & 3 \\ 4 & 2 & b \\ c & 7 & 5 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 4 & 3 \\ 4 & 2 & 7 \\ 3 & 7 & 5 \end{bmatrix}$$

Each unknown must equal its mirror entry across the diagonal: a is in position $(1, 2)$, whose mirror $(2, 1)$ holds 4, so $a = 4$; b is in position $(2, 3)$, whose mirror $(3, 2)$ holds 7, so $b = 7$; and c is in position $(3, 1)$, whose mirror $(1, 3)$ holds 3, so $c = 3$. The result, on the right, is determined by its diagonal and the three entries above it. This is why software often stores only half of a large symmetric matrix: the other half carries no new information.

Notebook U2-3_Matrices-1_ArithmeticAndSpecial, §3, the transpose

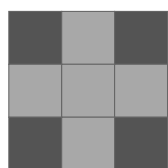
4.3 Application: An Image Is a Matrix

A grayscale image is a matrix of brightness values, one per pixel: 0 is black, and larger numbers are brighter. Here is a tiny “plus” sign, 3×3 pixels, stored as the matrix P . The plus is the brighter 2s running along the middle row and middle column; the corners are the dimmer 1s.

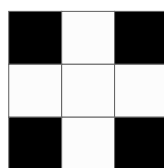
To boost the contrast, brighten the bright pixels and darken the dark ones: add 1 to every 2 and subtract 1 from every 1. That is **adding a matrix**:

$$P = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 2 & 2 \\ 1 & 2 & 1 \end{bmatrix}, \quad P + \begin{bmatrix} -1 & 1 & -1 \\ 1 & 1 & 1 \\ -1 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 3 & 0 \\ 3 & 3 & 3 \\ 0 & 3 & 0 \end{bmatrix}$$

Drawn as pixels, darker squares are smaller numbers:



original



higher contrast

After the boost the corners are pure black and the plus is brighter, so the shape stands out more sharply.

The other operations are photo adjustments too. Scaling by c dims the whole image (when $c < 1$) or brightens it (when $c > 1$), because every pixel's brightness is multiplied by the same factor. Transposing flips the image over its main diagonal; this particular plus sign is symmetric, $P^\top = P$, so it looks the same afterward. Every basic photo adjustment is matrix arithmetic, and a 4000×3000 photo is just a bigger grid.

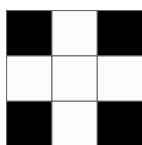
Example 4.12: Dimming. Scaling the original plus by $c = \frac{1}{2}$ halves every brightness:

$$\frac{1}{2}P = \begin{bmatrix} 0.5 & 1 & 0.5 \\ 1 & 1 & 1 \\ 0.5 & 1 & 0.5 \end{bmatrix}.$$

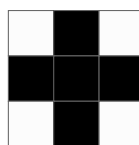
The picture keeps its shape (the plus is still twice as bright as the corners) but the whole image is darker. Brightness values need not be whole numbers; real image software typically stores values from 0 to 255, or from 0 to 1, and the same arithmetic applies.

Example 4.13: A photo negative. In a negative, bright becomes dark and dark becomes bright. With brightness running from 0 to 3, the negative of a pixel with value v is $3 - v$. For the high-contrast plus $Q = \begin{bmatrix} 0 & 3 & 0 \\ 3 & 3 & 3 \\ 0 & 3 & 0 \end{bmatrix}$, subtract Q from the matrix whose entries are all 3:

$$\begin{bmatrix} 3 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{bmatrix} - \begin{bmatrix} 0 & 3 & 0 \\ 3 & 3 & 3 \\ 0 & 3 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 3 \\ 0 & 0 & 0 \\ 3 & 0 & 3 \end{bmatrix}.$$



Q

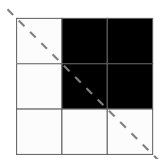


negative

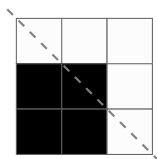
The bright plus on black corners has become a black plus on bright corners. A negative is one subtraction, applied to every pixel at once.

Example 4.14: Transposing an image that is not symmetric. The plus sign looked the same after transposing because it is symmetric. An “L” shape is not:

$$L = \begin{bmatrix} 3 & 0 & 0 \\ 3 & 0 & 0 \\ 3 & 3 & 3 \end{bmatrix}, \quad L^\top = \begin{bmatrix} 3 & 3 & 3 \\ 0 & 0 & 3 \\ 0 & 0 & 3 \end{bmatrix}.$$



L



$L^\top$

The first column of L (the upright stroke) has become the first row of $L^\top$, and the bottom row (the foot) has become the last column. The dashed line is the main diagonal: $L^\top$ is L reflected across it. A transpose is a genuine flip of the picture, which is why it changes any image that is not symmetric.

Notebook U2-3_Matrices-1_ArithmeticAndSpecial, §5, the plus sign drawn with imshow

4.4 Exercise: Arithmetic and Transposes

Exercise 4.1

Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & -1 \\ 2 & 5 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 0 \end{bmatrix}.$$

1. Compute $A + B$ and $3A - B$.
2. Write down $A^\top$ and $C^\top$, with their dimensions.
3. Compute $A + A^\top$. Is it symmetric? Would that be true for **any** square A ?
4. What is wrong with $A + C$?

Solution to Exercise 4.1

1. Add matching entries:

$$A + B = \begin{bmatrix} 1+0 & 2+(-1) \\ 3+2 & 4+5 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 5 & 9 \end{bmatrix}.$$

For $3A - B$, scale first, then subtract entry by entry:

$$3A = \begin{bmatrix} 3 & 6 \\ 9 & 12 \end{bmatrix}, \quad 3A - B = \begin{bmatrix} 3-0 & 6-(-1) \\ 9-2 & 12-5 \end{bmatrix} = \begin{bmatrix} 3 & 7 \\ 7 & 7 \end{bmatrix}.$$

2. Turn each row into a column:

$$A^\top = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}, \quad C^\top = \begin{bmatrix} 1 & 3 \\ 0 & 1 \\ 2 & 0 \end{bmatrix}.$$

$A^\top$ is still 2×2 , since a square matrix keeps its shape. $C^\top$ is now 3×2 : the two rows of C have become the two columns of $C^\top$.

3. Using $A^\top$ from part 2,

$$A + A^\top = \begin{bmatrix} 1+1 & 2+3 \\ 3+2 & 4+4 \end{bmatrix} = \begin{bmatrix} 2 & 5 \\ 5 & 8 \end{bmatrix},$$

which **is** symmetric: the two entries off the diagonal are both 5. And it always is, for any square A . Using the transpose rules and then the fact that addition does not care about order,

$$(A + A^\top)^\top = A^\top + (A^\top)^\top = A^\top + A = A + A^\top.$$

So $A + A^\top$ equals its own transpose. In words: the entry in position (i, j) is $a_{ij} + a_{ji}$, and the entry in position (j, i) is $a_{ji} + a_{ij}$, the same two numbers added in the other order. Adding a square matrix to its own transpose is the standard way to **manufacture** a symmetric matrix. (It has to be square: otherwise A and $A^\top$ have different shapes and cannot be added.)

4. **The shapes differ.** A is 2×2 and C is 2×3 , so there is no third column of A to add to the third column of C . The sum is undefined.

4.5 Extra Practice

Exercise 4.2

Let $A = \begin{bmatrix} 2 & 0 & -1 \\ 1 & 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 0 & -2 \\ 5 & 3 \end{bmatrix}$. For each of $A + B$, $A + B^\top$, $A^\top + B$, and $2A - B^\top$, say whether it is defined, and compute it if it is.

Solution to Exercise 4.2

Start with the shapes: A is 2×3 , B is 3×2 , so $B^\top$ is 2×3 and $A^\top$ is 3×2 .

- $A + B$: $(2 \times 3) + (3 \times 2)$, shapes differ, **undefined**.
- $A + B^\top$: both 2×3 , defined. With $B^\top = \begin{bmatrix} 1 & 0 & 5 \\ 1 & -2 & 3 \end{bmatrix}$,

$$A + B^\top = \begin{bmatrix} 2+1 & 0+0 & -1+5 \\ 1+1 & 3+(-2) & 4+3 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 4 \\ 2 & 1 & 7 \end{bmatrix}.$$

- $A^\top + B$: both 3×2 , defined. With $A^\top = \begin{bmatrix} 2 & 1 \\ 0 & 3 \\ -1 & 4 \end{bmatrix}$,

$$A^\top + B = \begin{bmatrix} 3 & 2 \\ 0 & 1 \\ 4 & 7 \end{bmatrix},$$

which is the transpose of the previous answer, as the rule $(A + B^\top)^\top = A^\top + B$ predicts.

- $2A - B^\top$: both 2×3 , defined.

$$2A - B^\top = \begin{bmatrix} 4-1 & 0-0 & -2-5 \\ 2-1 & 6-(-2) & 8-3 \end{bmatrix} = \begin{bmatrix} 3 & 0 & -7 \\ 1 & 8 & 5 \end{bmatrix}.$$

The lesson: check shapes before computing anything, and a transpose can turn an impossible sum into a possible one.

Exercise 4.3

Solve $3X - A = 2B$ for the matrix X , where $A = \begin{bmatrix} 3 & 0 \\ -3 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 3 \\ 3 & 0 \end{bmatrix}$, and check your answer.

Solution to Exercise 4.3

Add A to both sides, then scale by $\frac{1}{3}$: $X = \frac{1}{3}(A + 2B)$. Compute the pieces:

$$2B = \begin{bmatrix} 0 & 6 \\ 6 & 0 \end{bmatrix}, \quad A + 2B = \begin{bmatrix} 3 & 6 \\ 3 & 6 \end{bmatrix}, \quad X = \frac{1}{3} \begin{bmatrix} 3 & 6 \\ 3 & 6 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}.$$

Check: $3X - A = \begin{bmatrix} 3 & 6 \\ 3 & 6 \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ -3 & 6 \end{bmatrix} = \begin{bmatrix} 0 & 6 \\ 6 & 0 \end{bmatrix} = 2B$. ✓ The algebra is exactly the algebra of $3x - a = 2b$ for numbers, because addition and scalar multiplication obey the same rules.

Exercise 4.4

1. Find x , y , and z so that this matrix is symmetric:

$$\begin{bmatrix} 2 & x & 1 \\ 4 & 0 & y \\ z & -3 & 5 \end{bmatrix}.$$

2. True or false: every diagonal matrix is symmetric. Every symmetric matrix is diagonal.

Solution to Exercise 4.4

1. Match mirror pairs: x at $(1, 2)$ must equal the 4 at $(2, 1)$; y at $(2, 3)$ must equal the -3 at $(3, 2)$; z at $(3, 1)$ must equal the 1 at $(1, 3)$. So $x = 4$, $y = -3$, $z = 1$.
2. The first is **true**: every off-diagonal entry of a diagonal matrix is 0, so every mirror pair is $0 = 0$. The second is **false**: $\begin{bmatrix} 1 & 4 \\ 4 & 3 \end{bmatrix}$ is symmetric but has nonzero entries off the diagonal. Symmetric only asks that the off-diagonal entries *match* their mirrors, not that they vanish.

Exercise 4.5

A 2×3 image with brightness from 0 (black) to 3 (white) is $M = \begin{bmatrix} 0 & 1 & 2 \\ 3 & 2 & 1 \end{bmatrix}$. Compute its negative and its transpose, and say what shape each one has.

Solution to Exercise 4.5

The negative replaces each v by $3 - v$: subtract M from the 2×3 matrix of all 3s,

$$\begin{bmatrix} 3 & 3 & 3 \\ 3 & 3 & 3 \end{bmatrix} - \begin{bmatrix} 0 & 1 & 2 \\ 3 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix},$$

still 2×3 : a negative changes brightness, not layout. The transpose is

$$M^T = \begin{bmatrix} 0 & 3 \\ 1 & 2 \\ 2 & 1 \end{bmatrix},$$

which is 3×2 : an image 3 pixels wide and 2 tall has become 2 wide and 3 tall. Transposing a landscape photo gives a portrait one (flipped across its diagonal, not rotated).

5 Matrix Times Vector

5.1 Where the Product Comes From

In Unit 2-2 we asked how much of each of the basis vectors $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$ it takes to reach the point $(3, 3)$. We wrote that question as a combination and, comparing entries, as a system:

$$c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix} \quad \longleftrightarrow \quad \begin{cases} 2c_1 - c_2 = 3 \\ c_1 + c_2 = 3 \end{cases}$$

Both forms are clumsy: the unknowns c_1 and c_2 are scattered across the page, mixed in among the numbers. Here is the tidier idea. Put the basis vectors side by side as the **columns** of a matrix B , stack the unknowns into a vector $\bar{c}$, and **define** the product $B\bar{c}$ to mean exactly that combination:

$$\underbrace{\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}}_B \underbrace{\begin{bmatrix} c_1 \\ c_2 \end{bmatrix}}_{\bar{c}} = c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

The first entry of $\bar{c}$ multiplies the first column of B , the second entry multiplies the second column, and the results are added.

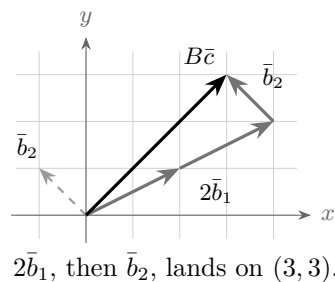
The idea

A matrix times a vector is the **linear combination of its columns**, weighted by the vector's entries.

Nothing new is being computed here; the combination is the same one as before. What is new is the notation. The product was invented so that “solve for the amounts” fits on one line: $B\bar{c} = \bar{p}$, where $\bar{p} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$ is the vector to the point $(3, 3)$. The matrix holds what is known (the basis), the vector $\bar{c}$ holds what is unknown (the amounts), and the right-hand side holds the target.

For completeness, here are the amounts. Adding the two equations cancels c_2 and leaves $3c_1 = 6$, so $c_1 = 2$, and then $c_2 = 3 - c_1 = 1$. So $B\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ should be $\bar{p}$, and by the idea above it is: two of the first column plus one of the second.

$$B \begin{bmatrix} 2 \\ 1 \end{bmatrix} = 2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 1 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 - 1 \\ 2 + 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}.$$



In the picture, $\bar{b}_1$ and $\bar{b}_2$ are the columns of B . The product $B\bar{c}$ is the route that walks c_1 copies of the first column and then c_2 copies of the second.

Example 5.1: Supply kits: a combination of columns. A clinic assembles three kinds of supply kit. Each kit is a vector listing how many gloves, masks, and swabs it contains, and the

three kits sit side by side as the columns of a matrix K :

$$K = \begin{bmatrix} 10 & 20 & 5 \\ 5 & 10 & 0 \\ 2 & 0 & 8 \end{bmatrix} \quad \begin{array}{l} \leftarrow \text{gloves} \\ \leftarrow \text{masks} \\ \leftarrow \text{swabs} \end{array} \quad (\text{columns: kit A, kit B, kit C}).$$

An order of 3 kit A's, 2 kit B's, and 4 kit C's is the vector $\bar{x} = \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix}$. The total supplies are three copies of column A, plus two of column B, plus four of column C, which is exactly $K\bar{x}$:

$$K\bar{x} = 3 \begin{bmatrix} 10 \\ 5 \\ 2 \end{bmatrix} + 2 \begin{bmatrix} 20 \\ 10 \\ 0 \end{bmatrix} + 4 \begin{bmatrix} 5 \\ 0 \\ 8 \end{bmatrix} = \begin{bmatrix} 30 \\ 15 \\ 6 \end{bmatrix} + \begin{bmatrix} 40 \\ 20 \\ 0 \end{bmatrix} + \begin{bmatrix} 20 \\ 0 \\ 32 \end{bmatrix} = \begin{bmatrix} 90 \\ 35 \\ 38 \end{bmatrix}.$$

The order uses 90 gloves, 35 masks, and 38 swabs. Whenever the columns of a matrix are “recipes” and the vector says how many of each recipe to use, the product is the combined result, and the column reading is the natural one.

5.2 Definition, and Its Two Readings

The idea above says what the product *means*. The official definition says how to compute it one entry at a time.

Definition: Matrix times vector

For an $m \times n$ matrix A and a vector $\bar{x} \in \mathbb{R}^n$, the product $A\bar{x}$ is the vector in $\mathbb{R}^m$ whose i th entry is **row i of A dotted with $\bar{x}$** :

$$(A\bar{x})_i = a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n = \sum_{j=1}^n a_{ij}x_j.$$

The symbol $\sum_{j=1}^n$ is shorthand for “add up the following terms as j runs from 1 to n .” Row i of A has n entries, a_{i1} through a_{in} , and $\bar{x}$ has n entries, x_1 through x_n ; pair them up, multiply each pair, and add.

Why do the definition and the idea agree? Take the combination $x_1(\text{column } 1) + \cdots + x_n(\text{column } n)$ and look only at its i th entry. Column j contributes its i th entry, a_{ij} , times x_j , so the i th entry of the combination is $a_{i1}x_1 + \cdots + a_{in}x_n$: exactly row i of A dotted with $\bar{x}$.

In the kit example, the row reading asks a different but equally natural question: row 1 of K is the glove count of each kit, $\begin{bmatrix} 10 & 20 & 5 \end{bmatrix}$, and dotting it with the order gives the total gloves, $10(3) + 20(2) + 5(4) = 30 + 40 + 20 = 90$. The column reading built the answer kit by kit; the row reading builds it supply by supply.

Worked example. We compute $A\bar{x}$ both ways for

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \bar{x} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}.$$

Reading 1: rows. Each entry of the answer is a dot product of a row of A with $\bar{x}$:

$$A\bar{x} = \begin{bmatrix} 1 \cdot 5 + 2 \cdot 6 \\ 3 \cdot 5 + 4 \cdot 6 \end{bmatrix} = \begin{bmatrix} 17 \\ 39 \end{bmatrix}.$$

Reading 2: columns. The answer is a combination of the columns of A , weighted by the entries of $\bar{x}$:

$$A\bar{x} = 5 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 6 \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 17 \\ 39 \end{bmatrix}.$$

It is the same arithmetic telling two stories: the same four products $1 \cdot 5$, $2 \cdot 6$, $3 \cdot 5$, $4 \cdot 6$ appear in both, just grouped differently. The row reading is how you **compute**; the column reading is how you **think**. In particular, the column reading says that $A\bar{x}$ always lives in the span of A 's columns, since it is a combination of them.

Shapes: $(m \times n)(n \times 1) = (m \times 1)$. The inner dimensions must match: each row of A has n entries, and $\bar{x}$ must also have n entries for the dot product to pair them up. The outer dimensions give the shape of the answer, one entry per row of A . So a 2×2 matrix cannot multiply a vector in $\mathbb{R}^3$: its rows have two entries and the vector has three.

outer sizes give the answer's shape

$$\begin{array}{ccc} \overbrace{(m \times n) \quad (n \times 1)} & = & (m \times 1) \\ \underbrace{\hspace{1.5cm}} & & \\ \text{inner sizes must match} & & \end{array}$$

Now three more products, each done both ways, in increasing order of difficulty.

Example 5.2: Negative entries. Compute $A\bar{x}$ for $A = \begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix}$ and $\bar{x} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$. The shapes are $(2 \times 2)(2 \times 1)$, so the answer is in $\mathbb{R}^2$.

$$\text{Rows: } A\bar{x} = \begin{bmatrix} 3(2) + (-1)(5) \\ 2(2) + 4(5) \end{bmatrix} = \begin{bmatrix} 6 - 5 \\ 4 + 20 \end{bmatrix} = \begin{bmatrix} 1 \\ 24 \end{bmatrix}.$$

$$\text{Columns: } 2 \begin{bmatrix} 3 \\ 2 \end{bmatrix} + 5 \begin{bmatrix} -1 \\ 4 \end{bmatrix} = \begin{bmatrix} 6 - 5 \\ 4 + 20 \end{bmatrix} = \begin{bmatrix} 1 \\ 24 \end{bmatrix}.$$

The middle steps are identical, term for term. When the two readings disagree in your own work, an arithmetic slip has happened, so computing both ways is a built-in check.

Example 5.3: A tall matrix: from $\mathbb{R}^2$ to $\mathbb{R}^3$. Compute $A\bar{x}$ for

$$A = \begin{bmatrix} 1 & 2 \\ 0 & -1 \\ 4 & 3 \end{bmatrix}, \quad \bar{x} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}.$$

The shapes are $(3 \times 2)(2 \times 1)$: inner sizes 2 and 2 match, and the answer is 3×1 . So a vector with two entries goes in and a vector with *three* comes out, one entry per row of A .

$$\text{Rows: } \begin{bmatrix} 1(3) + 2(-2) \\ 0(3) + (-1)(-2) \\ 4(3) + 3(-2) \end{bmatrix} = \begin{bmatrix} 3 - 4 \\ 0 + 2 \\ 12 - 6 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 6 \end{bmatrix}.$$

$$\text{Columns: } 3 \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix} - 2 \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 6 \end{bmatrix}.$$

In words: the output is a combination of two vectors in $\mathbb{R}^3$, so every possible output of this A lies on the plane those two columns span.

Example 5.4: A wide matrix: from $\mathbb{R}^3$ to $\mathbb{R}^2$. Compute $A\bar{x}$ for

$$A = \begin{bmatrix} 2 & 0 & -1 \\ 1 & 3 & 2 \end{bmatrix}, \quad \bar{x} = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}.$$

Now $(2 \times 3)(3 \times 1) = 2 \times 1$: three entries in, two out.

$$\text{Rows: } \begin{bmatrix} 2(4) + 0(1) + (-1)(2) \\ 1(4) + 3(1) + 2(2) \end{bmatrix} = \begin{bmatrix} 8 + 0 - 2 \\ 4 + 3 + 4 \end{bmatrix} = \begin{bmatrix} 6 \\ 11 \end{bmatrix}.$$

$$\text{Columns: } 4 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 1 \begin{bmatrix} 0 \\ 3 \end{bmatrix} + 2 \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 11 \end{bmatrix}.$$

Each row of A has three entries to pair with the three entries of $\bar{x}$, and there are two rows, so two dot products. The number of *rows* of A sets the size of the answer; the number of *columns* must match the size of the input.

Two data examples show why this product is everywhere in data science: applied to a data table, it computes a weighted score for *every row at once*.

Example 5.5: Course grades as a weighted score. Two students' scores on two midterms and a final are the rows of S , and the course weights the three exams 20%, 30%, and 50%:

$$S = \begin{bmatrix} 80 & 90 & 70 \\ 60 & 75 & 95 \end{bmatrix}, \quad \bar{w} = \begin{bmatrix} 0.2 \\ 0.3 \\ 0.5 \end{bmatrix}.$$

Each student's grade is their row dotted with the weights, which is the row reading of $S\bar{w}$:

$$S\bar{w} = \begin{bmatrix} 0.2(80) + 0.3(90) + 0.5(70) \\ 0.2(60) + 0.3(75) + 0.5(95) \end{bmatrix} = \begin{bmatrix} 16 + 27 + 35 \\ 12 + 22.5 + 47.5 \end{bmatrix} = \begin{bmatrix} 78 \\ 82 \end{bmatrix}.$$

Student 2 finishes ahead despite lower midterm scores, because the heavily weighted final pulled their grade up. The shapes, $(2 \times 3)(3 \times 1) = 2 \times 1$, say it all: one weight per exam goes in, one grade per student comes out. A class of 200 would be a 200×3 matrix and the same single product.

Example 5.6: A spam score for every email. Return to the word-count matrix E (rows are emails; columns are *free*, *win*, *cash*, *meeting*). A simple filter gives each word a weight, positive for spammy words and negative for ordinary ones, say $\bar{w} = \begin{bmatrix} 1 \\ 2 \\ 1 \\ -3 \end{bmatrix}$. Each email's score is its row of counts dotted with the weights:

$$E\bar{w} = \begin{bmatrix} 2 & 1 & 2 & 0 \\ 4 & 2 & 4 & 0 \\ 2 & 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \\ -3 \end{bmatrix} = \begin{bmatrix} 2 + 2 + 2 + 0 \\ 4 + 4 + 4 + 0 \\ 2 + 0 + 1 - 6 \end{bmatrix} = \begin{bmatrix} 6 \\ 12 \\ -3 \end{bmatrix}.$$

The two spam emails score high and the meeting email scores below zero, so a rule like “flag anything above 0” sorts all three correctly. The column reading tells the same story word by word:

$E\bar{w}$ is 1 times the *free* column, plus 2 times the *win* column, plus 1 times the *cash* column, minus 3 times the *meeting* column, so every word pushes every email's score up or down in proportion to how often it appears. This is exactly the shape of a **linear model**: a data matrix times a vector of weights gives one prediction per row. Linear regression makes its predictions the same way.

Watch out: dot the rows of A , not the columns

The row reading pairs $\bar{x}$ with the **rows** of A . Pairing it with the columns instead is a common slip, and it silently computes a different product. For $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $\bar{x} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$, dotting the *columns* with $\bar{x}$ gives $\begin{bmatrix} 1(5)+3(6) \\ 2(5)+4(6) \end{bmatrix} = \begin{bmatrix} 23 \\ 34 \end{bmatrix}$, not the correct $\begin{bmatrix} 17 \\ 39 \end{bmatrix}$. (What it actually computes is $A^\top \bar{x}$, since the columns of A are the rows of $A^\top$.) Memory aid: in the column reading the entries of $\bar{x}$ *scale* whole columns; in the row reading they *pair up* with the entries across a row.

Notebook: U2-3_Matrices-2_MultiplicationAndPowers, §1, both readings computed

5.3 Side by Side Now Means Something

In Unit 2-2, writing two symbols side by side had no meaning. Now it does, but only when the shapes fit. Here is the notation, updated:

Written	Shapes	Meaning
$A\bar{x}$	$(m \times n)(n \times 1)$	matrix times vector: a vector in $\mathbb{R}^m$
$\bar{x}A$	$(n \times 1)(m \times n)$	undefined : the inner sizes, 1 and m , do not match
$\bar{v} \bar{w}$	$(n \times 1)(n \times 1)$	still undefined for two columns
$\bar{v} \cdot \bar{w}$	two vectors	the dot product, a number

Order matters: $A\bar{x}$ is fine, but $\bar{x}A$ puts a single column in front of a matrix, and the inner sizes do not line up. Likewise two columns placed side by side cannot be multiplied. (Both “undefined” rows assume the matrix has more than one row and the vectors more than one entry, which is always the case in this course.) The dot product keeps its own symbol, the dot.

In fact the dot product *is* a matrix product, once one of the two vectors is tipped over. A row $\bar{v}^\top$ is a column tipped over, a $1 \times n$ matrix, and a row times a column has shape $(1 \times n)(n \times 1) = 1 \times 1$: a single number.

The dot product is a matrix product

$$\bar{v}^\top \bar{w} = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = v_1 w_1 + v_2 w_2 + v_3 w_3 = \bar{v} \cdot \bar{w}$$

For example,

$$\begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = 1 \cdot 3 + 2 \cdot 4 = 3 + 8 = 11.$$

And each entry of $A\bar{x}$ is one of these: a row of A times $\bar{x}$. That is exactly what the definition of $A\bar{x}$ said.

Example 5.7: Row times column, twice. First with three entries:

$$\begin{bmatrix} 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 1 \end{bmatrix} = 2(4) + (-1)(5) + 3(1) = 8 - 5 + 3 = 6.$$

The shapes are $(1 \times 3)(3 \times 1) = 1 \times 1$, a single number, and it is the dot product of $\begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$ with $\begin{bmatrix} 4 \\ 5 \\ 1 \end{bmatrix}$. Now a useful special case. A row of 1s times any column adds up the column's entries, so the three patients' ages from Section 3 total

$$\begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 54 \\ 58 \\ 41 \end{bmatrix} = 54 + 58 + 41 = 153, \quad \text{and} \quad \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 54 \\ 58 \\ 41 \end{bmatrix} = 51$$

is their mean age. Sums and averages, the most basic statistics there are, are row-times-column products; that is one reason statistical formulas are so often written with matrices.

5.4 A System Is Now One Equation

Back to the matinee. The augmented matrix of Unit 2-1 was this product in disguise:

$$\begin{cases} c + a + s = 300 \\ 6c + 10a + 8s = 2540 \\ -c + a - s = 0 \end{cases} \longleftrightarrow \underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 6 & 10 & 8 \\ -1 & 1 & -1 \end{bmatrix}}_A \underbrace{\begin{bmatrix} c \\ a \\ s \end{bmatrix}}_{\bar{x}} = \underbrace{\begin{bmatrix} 300 \\ 2540 \\ 0 \end{bmatrix}}_{\bar{b}}$$

To see why the two sides say the same thing, compute $A\bar{x}$ by the row reading. Row 1 of A dotted with $\bar{x}$ is $c + a + s$, row 2 gives $6c + 10a + 8s$, and row 3 gives $-c + a - s$. Setting that vector equal to $\bar{b}$, one entry at a time, gives back the three equations. So $A\bar{x} = \bar{b}$ is the system, packed into one line.

The augmented matrix $[A \mid \bar{b}]$ from Unit 2-1 is just A with $\bar{b}$ pinned on as an extra column.

Solve it in one line. With A and $\bar{b}$ stored as arrays, `np.linalg.solve(A, b)` gives $c = 80$, $a = 150$, $s = 70$: the theater sold 80 children's, 150 adults', and 70 seniors' tickets.

Check it with the row reading. Multiply A by the answer and confirm that $\bar{b}$ comes out:

$$A \begin{bmatrix} 80 \\ 150 \\ 70 \end{bmatrix} = \begin{bmatrix} 80 + 150 + 70 \\ 480 + 1500 + 560 \\ -80 + 150 - 70 \end{bmatrix} = \begin{bmatrix} 300 \\ 2540 \\ 0 \end{bmatrix} = \bar{b}.$$

Each row checks one fact from the story: 300 tickets in all, \$2540 taken in, and as many adult tickets as children's and seniors' combined ($150 = 80 + 70$).

The same check works for the vaccine clinic of Section 2:

$$\begin{bmatrix} 1 & 1 \\ 30 & 50 \end{bmatrix} \begin{bmatrix} 80 \\ 40 \end{bmatrix} = \begin{bmatrix} 80 + 40 \\ 2400 + 2000 \end{bmatrix} = \begin{bmatrix} 120 \\ 4400 \end{bmatrix},$$

the shot count and the billing total. Checking a proposed solution is always a single matrix-vector product.

Example 5.8: From $A\bar{x} = \bar{b}$ back to equations, and checking candidates. Let

$$A = \begin{bmatrix} 2 & -1 & 0 \\ 1 & 1 & 1 \\ 0 & 3 & -2 \end{bmatrix}, \quad \bar{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad \bar{b} = \begin{bmatrix} 1 \\ 6 \\ -5 \end{bmatrix}.$$

Reading $A\bar{x} = \bar{b}$ row by row gives the system

$$\begin{cases} 2x - y = 1 \\ x + y + z = 6 \\ 3y - 2z = -5 \end{cases}$$

(the zeros in A are the missing variables). Is $\bar{x} = \begin{bmatrix} 1 \\ 1 \\ 4 \end{bmatrix}$ a solution? Is $\begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$?

$$A \begin{bmatrix} 1 \\ 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 2 - 1 + 0 \\ 1 + 1 + 4 \\ 0 + 3 - 8 \end{bmatrix} = \begin{bmatrix} 1 \\ 6 \\ -5 \end{bmatrix} = \bar{b} \checkmark, \quad A \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 - 3 + 0 \\ 2 + 3 + 1 \\ 0 + 9 - 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 6 \\ 7 \end{bmatrix} \neq \bar{b}.$$

The first candidate solves the system. The second satisfies the first two equations but misses the third (7 instead of -5), so it is not a solution: a solution must satisfy *every* row. The product shows at a glance which equations a candidate passes and which it fails.

Example 5.9: From equations to $A\bar{x} = \bar{b}$, carefully. Write each system as $A\bar{x} = \bar{b}$.

1. $x + 2z = 5$ and $3y = 6$. There are two equations and three unknowns, so A is 2×3 . Every unknown gets a column even where it is missing, with a 0 there:

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 3 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}.$$

2. $2x + y = 5$ and $y - x = 2$. The second equation lists y first, but the columns must keep one order throughout, x then y . Rewrite it as $-x + y = 2$:

$$\begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}.$$

Substituting $y = x + 2$ into the first equation gives $3x + 2 = 5$, so $x = 1$ and $y = 3$, and the product confirms it:

$$\begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 + 3 \\ -1 + 3 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}.$$

Had we copied the second row in the order written, as $\begin{bmatrix} 1 & -1 \end{bmatrix}$, the same check would give $\begin{bmatrix} 5 \\ -2 \end{bmatrix}$ and expose the mistake.

Watch out: one column per unknown, in one fixed order

When turning equations into $A\bar{x} = \bar{b}$: (1) move every unknown to the left side and every constant to the right; (2) choose an order for the unknowns and use it in *every* row; (3) write

0 for any unknown an equation leaves out. Skipping step 3 shifts the later coefficients into the wrong columns, and skipping step 2 silently swaps coefficients. Both mistakes produce a perfectly good-looking matrix for the wrong problem.

Example 5.10: Forward and backward with the supply kits. Example 5.1 ran the kit problem *forward*: given the order $\bar{x}$, compute the supplies $K\bar{x}$. Often the question runs *backward*: the clinic needs exactly 90 gloves, 35 masks, and 38 swabs; what order provides them? That is solving $K\bar{x} = \begin{bmatrix} 90 \\ 35 \\ 38 \end{bmatrix}$ for $\bar{x}$, a system of three equations, and `np.linalg.solve(K, b)` returns $\bar{x} = \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix}$, the order we started from. The two directions are the two jobs this notation does: **multiplying** computes an outcome from known inputs, and **solving** recovers the inputs from a desired outcome.

5.5 What the Special Matrices Do

Now that we can multiply a matrix by a vector, the special matrices of Section 3 earn their names.

The identity does nothing.

$$I\bar{x} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1x + 0y \\ 0x + 1y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} = \bar{x}$$

Each row of I picks out one entry, unchanged: the 1 selects it and the 0 ignores the other. The identity is to matrices what the number 1 is to multiplication, which is where its name comes from.

A diagonal matrix scales each coordinate by its own factor.

$$\begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x + 0y \\ 0x + 5y \end{bmatrix} = \begin{bmatrix} 2x \\ 5y \end{bmatrix}$$

The first coordinate doubles and the second is multiplied by 5. The zeros off the diagonal are what keep the two coordinates from mixing. When every diagonal factor is the same, the matrix stretches the whole vector evenly: $D = 4I$ from Exercise 3.1 multiplies both coordinates by 4, stretching everything by 4.

Example 5.11: The identity and the zero matrix in three dimensions.

$$I_3 \begin{bmatrix} 4 \\ -2 \\ 7 \end{bmatrix} = \begin{bmatrix} 1(4) + 0(-2) + 0(7) \\ 0(4) + 1(-2) + 0(7) \\ 0(4) + 0(-2) + 1(7) \end{bmatrix} = \begin{bmatrix} 4 \\ -2 \\ 7 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ -2 \\ 7 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Row k of I_3 has a single 1, in position k , so it picks out the k th entry of the vector and nothing else; the vector comes back unchanged. The zero matrix is the opposite extreme: every row dotted with anything is 0, so every input is sent to a zero vector. (Here the zero matrix is 2×3 , so the zero vector that comes out lives in $\mathbb{R}^2$.)

Example 5.12: A diagonal matrix converts units, one coordinate at a time. A patient vector records systolic blood pressure in mmHg and fasting glucose in mg/dL, $\begin{bmatrix} 130 \\ 126 \end{bmatrix}$. To report

glucose in mmol/L while leaving blood pressure alone, divide the second coordinate by 18 and leave the first unchanged: a job for a diagonal matrix, with factors 1 and $\frac{1}{18}$.

$$\begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{18} \end{bmatrix} \begin{bmatrix} 130 \\ 126 \end{bmatrix} = \begin{bmatrix} 1(130) + 0(126) \\ 0(130) + \frac{1}{18}(126) \end{bmatrix} = \begin{bmatrix} 130 \\ 7 \end{bmatrix}.$$

A scalar could not do this (it would scale both coordinates by the same factor); a diagonal matrix gives each coordinate its own factor. This is how a pipeline rescales features that are in different units.

Example 5.13: A zero on the diagonal, and a matrix that mixes. A diagonal entry of 0 wipes out its coordinate:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 8 \\ -3 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ -3 \end{bmatrix}.$$

The first and third coordinates pass through and the second is switched off, as if that feature had been dropped. For contrast, here is a matrix that is *not* diagonal:

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2+3 \\ 0+3 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}.$$

The off-diagonal 1 lets the second coordinate leak into the first: the new first coordinate is the *sum* of the old two. Off-diagonal entries are exactly what mix coordinates together, and a diagonal matrix, having none, never mixes.

So a matrix A takes a vector **in** and hands a vector **out**. In the language of Unit 1-3, a matrix is a **function** on vectors. Unit 2-4 develops that point of view.

5.6 Exercise: Matrix Times Vector

Exercise 5.1

Let

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix} \quad \text{and} \quad \bar{u} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

1. Compute $A\bar{u}$ by the row reading, then again by the column reading.
2. Compute $A^T\bar{u}$. Is it the same as $A\bar{u}$?
3. Write the system $2x + 3y = 7$, $x + 4y = 9$ in the form $A\bar{x} = \bar{b}$.
4. Compute $I\bar{u}$ and $\begin{bmatrix} 3 & 0 \\ 0 & -1 \end{bmatrix}\bar{u}$, and say in words what each matrix did.

Solution to Exercise 5.1

1. **Rows:** dot each row of A with $\bar{u}$.

$$A\bar{u} = \begin{bmatrix} 2(2) + 3(-1) \\ 1(2) + 4(-1) \end{bmatrix} = \begin{bmatrix} 4 - 3 \\ 2 - 4 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}.$$

Columns: weight the first column of A by 2 and the second by -1 , and add.

$$2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} - 1 \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 4 - 3 \\ 2 - 4 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}.$$

They agree, as they must: the same four products appear in both, grouped differently.

2. First transpose, turning the rows of A into columns, then multiply:

$$A^T = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}, \quad A^T \bar{u} = \begin{bmatrix} 2(2) + 1(-1) \\ 3(2) + 4(-1) \end{bmatrix} = \begin{bmatrix} 4 - 1 \\ 6 - 4 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

This is **different** from $A\bar{u} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$. Transposing changes which numbers sit in each row, so it changes the product. (The two would agree only if A were symmetric; here $a_{12} = 3$ but $a_{21} = 1$.)

3. The coefficients of x and y in each equation form one row of the matrix:

$$\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 7 \\ 9 \end{bmatrix}.$$

The coefficient grid is A (the same A as above), the unknowns form $\bar{x} = \begin{bmatrix} x \\ y \end{bmatrix}$, and the right-hand sides form $\bar{b} = \begin{bmatrix} 7 \\ 9 \end{bmatrix}$. The row reading confirms it: row 1 gives $2x + 3y$ and row 2 gives $x + 4y$.

4. For the identity,

$$I\bar{u} = \begin{bmatrix} 1(2) + 0(-1) \\ 0(2) + 1(-1) \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \bar{u}.$$

The identity **left $\bar{u}$ alone**. For the diagonal matrix,

$$\begin{bmatrix} 3 & 0 \\ 0 & -1 \end{bmatrix} \bar{u} = \begin{bmatrix} 3(2) + 0(-1) \\ 0(2) + (-1)(-1) \end{bmatrix} = \begin{bmatrix} 6 \\ 1 \end{bmatrix}.$$

The diagonal matrix **tripled the first coordinate and flipped the sign of the second**, each on its own, with no mixing between them.

5.7 Extra Practice

Exercise 5.2

Compute $\begin{bmatrix} 1 & -2 & 0 \\ 3 & 1 & 2 \\ 0 & 4 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$ by the row reading and again by the column reading.

Solution to Exercise 5.2

The shapes are $(3 \times 3)(3 \times 1)$, so the answer is in $\mathbb{R}^3$. **Rows:**

$$\begin{bmatrix} 1(2) + (-2)(1) + 0(3) \\ 3(2) + 1(1) + 2(3) \\ 0(2) + 4(1) + (-1)(3) \end{bmatrix} = \begin{bmatrix} 2 - 2 + 0 \\ 6 + 1 + 6 \\ 0 + 4 - 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 13 \\ 1 \end{bmatrix}.$$

Columns:

$$2 \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} -2 \\ 1 \\ 4 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 - 2 + 0 \\ 6 + 1 + 6 \\ 0 + 4 - 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 13 \\ 1 \end{bmatrix}.$$

The two agree. Notice the 0 in the first entry: an entry of a product can be 0 through cancellation, even though the row and the vector involved are mostly nonzero; here the contributions 2 and -2 cancelled.

Exercise 5.3

Let A be a 2×3 matrix, $\bar{u} \in \mathbb{R}^3$, and $\bar{v} \in \mathbb{R}^2$. For each of $A\bar{u}$, $A\bar{v}$, $A^\top \bar{v}$, $A^\top \bar{u}$, $\bar{u}^\top \bar{u}$, and $\bar{u}^\top \bar{v}$, say whether it is defined and, if it is, what shape the result has.

Solution to Exercise 5.3

Write each as shapes and compare the inner sizes.

- $A\bar{u}$: $(2 \times 3)(3 \times 1)$, inner $3 = 3$. **Defined**, a vector in $\mathbb{R}^2$.
- $A\bar{v}$: $(2 \times 3)(2 \times 1)$, inner $3 \neq 2$. **Undefined**.
- $A^\top \bar{v}$: $A^\top$ is 3×2 , so $(3 \times 2)(2 \times 1)$, inner $2 = 2$. **Defined**, a vector in $\mathbb{R}^3$.
- $A^\top \bar{u}$: $(3 \times 2)(3 \times 1)$, inner $2 \neq 3$. **Undefined**.
- $\bar{u}^\top \bar{u}$: $(1 \times 3)(3 \times 1)$. **Defined**, a single number: $\bar{u} \cdot \bar{u}$, the squared length $\|\bar{u}\|^2$ of Unit 2-2.
- $\bar{u}^\top \bar{v}$: $(1 \times 3)(2 \times 1)$, inner $3 \neq 2$. **Undefined**, just as a dot product of vectors of different sizes was in Unit 2-2.

Transposing A swaps which vectors it can accept: A takes vectors in $\mathbb{R}^3$, and $A^\top$ takes vectors in $\mathbb{R}^2$.

Exercise 5.4

Using the word-count matrix E of Example 5.6, compute the scores $E\bar{w}$ for the gentler weights $\bar{w} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$. Does the rule “flag anything above 0” still sort the three emails correctly?

Solution to Exercise 5.4

Dot each row of E with the new weights:

$$E\bar{w} = \begin{bmatrix} 2 + 1 + 2 - 0 \\ 4 + 2 + 4 - 0 \\ 2 + 0 + 1 - 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \\ 1 \end{bmatrix}.$$

All three scores are now above 0, so the rule would flag the meeting email as spam too. With the weaker penalty on *meeting*, its two occurrences no longer outweigh the three spammy words in it. The same data with different weights gives different predictions; choosing the weights well is what *training* a model means.

Exercise 5.5

Write the system $3x - y + 2z = 9$, $x + 4z = 17$, $2y - z = 0$ as $A\bar{x} = \bar{b}$. Then use a matrix-vector product to test whether $\begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix}$ are solutions.

Solution to Exercise 5.5

With the unknowns in the order x, y, z , and a 0 for each missing one,

$$\underbrace{\begin{bmatrix} 3 & -1 & 2 \\ 1 & 0 & 4 \\ 0 & 2 & -1 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{\bar{b}} = \underbrace{\begin{bmatrix} 9 \\ 17 \\ 0 \end{bmatrix}}_{\bar{b}}.$$

Test the candidates:

$$A \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 - 2 + 8 \\ 1 + 0 + 16 \\ 0 + 4 - 4 \end{bmatrix} = \begin{bmatrix} 9 \\ 17 \\ 0 \end{bmatrix} = \bar{b} \checkmark, \quad A \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 9 - 2 + 8 \\ 3 + 0 + 16 \\ 0 + 4 - 4 \end{bmatrix} = \begin{bmatrix} 15 \\ 19 \\ 0 \end{bmatrix} \neq \bar{b}.$$

The first vector is a solution. The second satisfies only the third equation, so it is not.

Exercise 5.6

With the kit matrix K of Example 5.1, does the order “5 of kit A, 2 of kit B, 2 of kit C” supply exactly 100 gloves, 45 masks, and 26 swabs? Answer with one product, and say which reading you used.

Solution to Exercise 5.6

Compute $K\bar{x}$ for $\bar{x} = \begin{bmatrix} 5 \\ 2 \\ 2 \end{bmatrix}$. By the column reading (so many copies of each kit):

$$5 \begin{bmatrix} 10 \\ 5 \\ 2 \end{bmatrix} + 2 \begin{bmatrix} 20 \\ 10 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 5 \\ 0 \\ 8 \end{bmatrix} = \begin{bmatrix} 50 + 40 + 10 \\ 25 + 20 + 0 \\ 10 + 0 + 16 \end{bmatrix} = \begin{bmatrix} 100 \\ 45 \\ 26 \end{bmatrix}.$$

Yes, exactly. (The row reading gives the same numbers supply by supply; for example, gloves

are $\begin{bmatrix} 10 & 20 & 5 \end{bmatrix}$ dotted with the order, $50 + 40 + 10 = 100$.)

6 Matrix Times Matrix

6.1 The Row-by-Column Rule

A matrix times a vector took rows of the matrix and dotted them with the vector. A matrix times a matrix does the same thing once for every column of the second matrix.

Definition: Matrix times matrix

If A is $m \times n$ and B is $n \times p$, the product AB is the $m \times p$ matrix whose (i, j) entry is **row i of A dotted with column j of B** :

$$(AB)_{ij} = \sum_{k=1}^n a_{ik} b_{kj}.$$

In the sum, k walks along row i of A (through $a_{i1}, a_{i2}, \dots, a_{in}$) and, in step, down column j of B (through $b_{1j}, b_{2j}, \dots, b_{nj}$). Multiply the pairs and add.

Worked example. Every entry of a 2×2 product is one dot product. With $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$,

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 1 \cdot 5 + 2 \cdot 7 & 1 \cdot 6 + 2 \cdot 8 \\ 3 \cdot 5 + 4 \cdot 7 & 3 \cdot 6 + 4 \cdot 8 \end{bmatrix} = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}.$$

To find the entry in row i , column j , slide along row i of the left matrix and down column j of the right one, multiplying the pairs you meet and adding. For instance, the top-right 22 is row 1 of A against column 2 of B : $1 \cdot 6 + 2 \cdot 8 = 6 + 16 = 22$.

Shapes: $(m \times n)(n \times p) = (m \times p)$. A row of A has n entries, and it pairs with a column of B that also has n ; that is why the inner dimensions must match. The answer has one row for each row of A and one column for each column of B . The matrix-vector product is the case $p = 1$: a vector is a matrix with a single column.

A good habit, before computing a single entry: write the two shapes side by side, check that the inner numbers match, and write down the shape of the answer. Then fill it in one entry at a time, saying “row i of the left, column j of the right” for each.

Example 6.1: A product with negatives. Compute AB for $A = \begin{bmatrix} 2 & -1 \\ 0 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 4 \\ 5 & -2 \end{bmatrix}$. Shapes: $(2 \times 2)(2 \times 2) = 2 \times 2$. Four entries, four dot products:

$$\begin{aligned} (AB)_{11} &= \text{row } 1 \cdot \text{col } 1 = 2(1) + (-1)(5) = -3, & (AB)_{12} &= \text{row } 1 \cdot \text{col } 2 = 2(4) + (-1)(-2) = 10, \\ (AB)_{21} &= \text{row } 2 \cdot \text{col } 1 = 0(1) + 3(5) = 15, & (AB)_{22} &= \text{row } 2 \cdot \text{col } 2 = 0(4) + 3(-2) = -6, \end{aligned}$$

so

$$AB = \begin{bmatrix} -3 & 10 \\ 15 & -6 \end{bmatrix}.$$

The sign slips hide in entries like $(AB)_{12}$, where $(-1)(-2) = +2$. Labeling each entry by its row and column, as above, keeps the bookkeeping straight.

Example 6.2: Different shapes: a 2×3 times a 3×2 . Let

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1 \\ 0 & 3 \\ -1 & 5 \end{bmatrix}.$$

Shapes first: $(2 \times 3)(3 \times 2)$. The inner sizes are 3 and 3, so the product exists, and the outer sizes say AB is 2×2 . Each entry pairs a row of A (three entries) with a column of B (three entries):

$$\begin{aligned} (AB)_{11} &= 1(2) + 2(0) + 0(-1) = 2, & (AB)_{12} &= 1(1) + 2(3) + 0(5) = 7, \\ (AB)_{21} &= 3(2) + (-1)(0) + 4(-1) = 2, & (AB)_{22} &= 3(1) + (-1)(3) + 4(5) = 20. \end{aligned}$$

$$AB = \begin{bmatrix} 2 & 7 \\ 2 & 20 \end{bmatrix}.$$

Two 3-sized dimensions went in and disappeared: the “inner” size is used up by the dot products, and only the outer sizes survive. We will multiply these two in the other order in the subsection on rules.

Example 6.3: Rows and columns as matrices. Vectors are matrices too, so the same rule multiplies them, and the shapes decide what comes out. A 1×3 row times a 3×2 matrix is a 1×2 row:

$$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1+0+3 & 0+2+3 \end{bmatrix} = \begin{bmatrix} 4 & 5 \end{bmatrix}.$$

And a 3×1 column times a 1×2 row is a whole 3×2 matrix, because the inner size is 1 and each “dot product” has a single term:

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 4 & 5 \end{bmatrix} = \begin{bmatrix} 1 \cdot 4 & 1 \cdot 5 \\ 2 \cdot 4 & 2 \cdot 5 \\ 3 \cdot 4 & 3 \cdot 5 \end{bmatrix} = \begin{bmatrix} 4 & 5 \\ 8 & 10 \\ 12 & 15 \end{bmatrix}.$$

In words: every entry of the column multiplies every entry of the row, and the results fill a grid.

Watch out: $\bar{v}^\top \bar{w}$ and $\bar{v} \bar{w}^\top$ are very different

For two vectors in $\mathbb{R}^n$, the row-times-column product $\bar{v}^\top \bar{w}$ is $(1 \times n)(n \times 1) = 1 \times 1$: the dot product, a single number. The column-times-row product $\bar{v} \bar{w}^\top$ is $(n \times 1)(1 \times n) = n \times n$: a whole matrix. Same two vectors, same symbols, opposite order, and the answers are not even the same kind of object. Whenever a transpose appears, check the shapes before anything else.

Example 6.4: A shape drill. Suppose A is 2×3 , B is 3×4 , C is 4×2 , and D is 3×3 . Which products exist? For each, write the shapes side by side and compare the inner numbers.

Product	Shapes	Inner sizes	Result
AB	$(2 \times 3)(3 \times 4)$	$3 = 3$	2×4
BA	$(3 \times 4)(2 \times 3)$	$4 \neq 2$	undefined
BC	$(3 \times 4)(4 \times 2)$	$4 = 4$	3×2
CB	$(4 \times 2)(3 \times 4)$	$2 \neq 3$	undefined
CA	$(4 \times 2)(2 \times 3)$	$2 = 2$	4×3
AD	$(2 \times 3)(3 \times 3)$	$3 = 3$	2×3
DA	$(3 \times 3)(2 \times 3)$	$3 \neq 2$	undefined
DB	$(3 \times 3)(3 \times 4)$	$3 = 3$	3×4
ABC	$(2 \times 3)(3 \times 4)(4 \times 2)$	$3 = 3, 4 = 4$	2×2

For a chain like ABC , the shapes must link like dominoes, each inner pair matching; the answer takes the first matrix's row count and the last matrix's column count. Notice how often reversing the order breaks a product: AB exists but BA does not, and the same for BC and CB . This table is exactly what `numpy` checks before it multiplies, and a “shapes not aligned” error means one of these inner pairs failed.

Notebook U2-3_Matrices-2_MultiplicationAndPowers, §2, the top-left entry checked by hand

6.2 Why That Rule, and Not Entry by Entry

Multiplying matching entries, as we did for addition, would be easy, and it would be useless: it would not connect to the matrix-vector product at all. The row-by-column rule is chosen for a reason: so that AB means “do B , then do A .” The way to see this is to look at the product one column at a time.

The columns of a product

Write B by its columns, $B = [\bar{b}_1 \ \bar{b}_2]$. Then

$$AB = A[\bar{b}_1 \ \bar{b}_2] = [A\bar{b}_1 \ A\bar{b}_2].$$

Each column of AB is A times the corresponding column of B . A matrix-matrix product is several matrix-vector products done side by side.

This follows straight from the definition. Column j of AB is made of the entries $(AB)_{1j}, (AB)_{2j}, \dots$, each of which is a row of A dotted with column j of B . By the row reading, that list is exactly A times column j of B .

Check on the worked example.

$$A \begin{bmatrix} 5 \\ 7 \end{bmatrix} = \begin{bmatrix} 1 \cdot 5 + 2 \cdot 7 \\ 3 \cdot 5 + 4 \cdot 7 \end{bmatrix} = \begin{bmatrix} 19 \\ 43 \end{bmatrix}, \quad A \begin{bmatrix} 6 \\ 8 \end{bmatrix} = \begin{bmatrix} 1 \cdot 6 + 2 \cdot 8 \\ 3 \cdot 6 + 4 \cdot 8 \end{bmatrix} = \begin{bmatrix} 22 \\ 50 \end{bmatrix},$$

and these are the two columns of AB .

The same product can also be read by *rows*: row i of AB is row i of A times the whole matrix B , which is a combination of the rows of B weighted by the entries of that row of A . In Example 6.2,

row 1 of A is $\begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$, so row 1 of AB is

$$1 \begin{bmatrix} 2 & 1 \end{bmatrix} + 2 \begin{bmatrix} 0 & 3 \end{bmatrix} + 0 \begin{bmatrix} -1 & 5 \end{bmatrix} = \begin{bmatrix} 2 & 7 \end{bmatrix},$$

matching what we found entry by entry. Columns of AB are combinations of columns of A ; rows of AB are combinations of rows of B . Both views come from the one rule.

Now think of each matrix as a function on vectors, as in Section 5. Suppose B sends a vector $\bar{x}$ to $B\bar{x}$, and A then sends that to $A(B\bar{x})$. The single matrix AB does both steps at once:

$$(AB)\bar{x} = A(B\bar{x}).$$

Note the order: the matrix applied **first** sits on the **right**, next to $\bar{x}$. This is the same convention as composing functions in Unit 1-3, where $f(g(x))$ applies g first even though f is written first.

That is the reason for the rule, and Unit 2-4 will use it to chain a rotation and a reflection into one matrix.

Example 6.5: Two steps, one matrix: from kit orders to dollars. Return to the supply kits of Example 5.1. The kit matrix K turns an *order* (how many of each kit) into *supplies* (how many gloves, masks, swabs). Suppose, with made-up round prices, a glove costs \$1, a mask \$2, and a swab \$3. The price row $\bar{r}^\top = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$ turns *supplies* into *dollars*. So there are two steps:

$$\text{order } \bar{x} \xrightarrow{K} \text{supplies } K\bar{x} \xrightarrow{\bar{r}^\top} \text{dollars } \bar{r}^\top(K\bar{x}).$$

The product $\bar{r}^\top K$, shapes $(1 \times 3)(3 \times 3) = 1 \times 3$, does both steps at once. Its entries are the price row dotted with each kit's column:

$$\bar{r}^\top K = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 10 & 20 & 5 \\ 5 & 10 & 0 \\ 2 & 0 & 8 \end{bmatrix} = \begin{bmatrix} 10 + 10 + 6 & 20 + 20 + 0 & 5 + 0 + 24 \end{bmatrix} = \begin{bmatrix} 26 & 40 & 29 \end{bmatrix},$$

the cost of one kit A, one kit B, and one kit C. For the order $\bar{x} = \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix}$, the two routes agree:

$$(\bar{r}^\top K)\bar{x} = 26(3) + 40(2) + 29(4) = 78 + 80 + 116 = 274,$$

$$\bar{r}^\top(K\bar{x}) = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 90 \\ 35 \\ 38 \end{bmatrix} = 90 + 70 + 114 = 274.$$

The order costs \$274 either way. The product $\bar{r}^\top K$ is the useful object to keep: a per-kit price list that prices any future order in one dot product, with no need to count supplies first.

Watch out: in numpy, $*$ is not matrix multiplication

For arrays A and B , $A * B$ multiplies **matching entries**, the “easy and useless” rule above, while $A @ B$ is the matrix product. With the worked example’s matrices, $A * B$ gives $\begin{bmatrix} 5 & 12 \\ 21 & 32 \end{bmatrix}$ but $A @ B$ gives $\begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$. Both run without an error on two 2×2 arrays, so the wrong one fails silently. Use $@$ whenever you mean the product of these notes.

6.3 Application: A Bill in One Product

Two students take credits at two colleges that charge different rates. In the credits matrix, rows are students and columns are colleges: student 1 takes 10 credits at the first college and 7 at the second, and student 2 takes 11 and 4. The first college charges \$60 per credit and the second \$80. Row i dotted with the cost vector is student i 's bill:

$$\underbrace{\begin{bmatrix} 10 & 7 \\ 11 & 4 \end{bmatrix}}_{\text{credits}} \underbrace{\begin{bmatrix} 60 \\ 80 \end{bmatrix}}_{\text{cost per credit}} = \begin{bmatrix} 10(60) + 7(80) \\ 11(60) + 4(80) \end{bmatrix} = \underbrace{\begin{bmatrix} 1160 \\ 980 \end{bmatrix}}_{\text{each student's bill}}$$

Student 1 pays $600 + 560 = 1160$ dollars and student 2 pays $660 + 320 = 980$ dollars.

Now add a second scenario. Suppose both colleges plan a 10% increase, to \$66 and \$88 per credit. Put the two cost vectors side by side as the columns of one cost matrix; each column of the answer is then one scenario:

$$\begin{bmatrix} 10 & 7 \\ 11 & 4 \end{bmatrix} \begin{bmatrix} 60 & 66 \\ 80 & 88 \end{bmatrix} = \begin{bmatrix} 1160 & 1276 \\ 980 & 1078 \end{bmatrix}$$

Column 1 is the bill now, and column 2 is the bill after the increase: $10(66) + 7(88) = 660 + 616 = 1276$ and $11(66) + 4(88) = 726 + 352 = 1078$. This is the column rule doing real work: the credits matrix times each column of the cost matrix, with every scenario handled in a single product.

Example 6.6: Comparing two suppliers. Back to the clinic kits. A second supplier quotes \$2 per glove, \$1 per mask, and \$2 per swab. Stack the two suppliers' price rows into a 2×3 matrix R (one row per supplier) and multiply by the kit matrix:

$$RK = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 2 \end{bmatrix} \begin{bmatrix} 10 & 20 & 5 \\ 5 & 10 & 0 \\ 2 & 0 & 8 \end{bmatrix} = \begin{bmatrix} 10 + 10 + 6 & 20 + 20 + 0 & 5 + 0 + 24 \\ 20 + 5 + 4 & 40 + 10 + 0 & 10 + 0 + 16 \end{bmatrix} = \begin{bmatrix} 26 & 40 & 29 \\ 29 & 50 & 26 \end{bmatrix}.$$

Row 1 is supplier 1's per-kit price list (from Example 6.5) and row 2 is supplier 2's. Supplier 2 is cheaper for kit C but dearer for kits A and B. Which is cheaper for the order $\bar{x} = \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix}$?

$$(RK)\bar{x} = \begin{bmatrix} 26(3) + 40(2) + 29(4) \\ 29(3) + 50(2) + 26(4) \end{bmatrix} = \begin{bmatrix} 78 + 80 + 116 \\ 87 + 100 + 104 \end{bmatrix} = \begin{bmatrix} 274 \\ 291 \end{bmatrix}.$$

Supplier 1 wins this order, \$274 against \$291. As a check on the "two routes" idea, counting the supplies first gives the same totals:

$$R(K\bar{x}) = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 2 \end{bmatrix} \begin{bmatrix} 90 \\ 35 \\ 38 \end{bmatrix} = \begin{bmatrix} 90 + 70 + 114 \\ 180 + 35 + 76 \end{bmatrix} = \begin{bmatrix} 274 \\ 291 \end{bmatrix}.$$

One product compared every supplier on every kit; adding a third supplier would just add a row to R .

Example 6.7: A table of dot products: EE^T . Section 2 promised that products like $X^T X$, the heart of least-squares regression, would be within reach. Here is one of the same kind, built

from the email matrix E (rows are the emails $\bar{d}_1^\top, \bar{d}_2^\top, \bar{d}_3^\top$). Shapes: E is 3×4 and $E^\top$ is 4×3 , so $EE^\top$ is 3×3 . Its (i, j) entry is row i of E dotted with column j of $E^\top$, and column j of $E^\top$ is row j of E . So every entry is a dot product between two emails:

$$EE^\top = \begin{bmatrix} \bar{d}_1 \cdot \bar{d}_1 & \bar{d}_1 \cdot \bar{d}_2 & \bar{d}_1 \cdot \bar{d}_3 \\ \bar{d}_2 \cdot \bar{d}_1 & \bar{d}_2 \cdot \bar{d}_2 & \bar{d}_2 \cdot \bar{d}_3 \\ \bar{d}_3 \cdot \bar{d}_1 & \bar{d}_3 \cdot \bar{d}_2 & \bar{d}_3 \cdot \bar{d}_3 \end{bmatrix} = \begin{bmatrix} 9 & 18 & 6 \\ 18 & 36 & 12 \\ 6 & 12 & 9 \end{bmatrix}.$$

For instance, $\bar{d}_1 \cdot \bar{d}_3 = 2(2) + 1(0) + 2(1) + 0(2) = 6$. What the table says:

- The **diagonal** holds each email dotted with itself, its squared length: 9, 36, 9, so the lengths are 3, 6, 3, as Unit 2-2 found.
- The **off-diagonal** entries are the dot products between different emails. Divide each by the two lengths and you get the cosine similarities, $18/(3 \cdot 6) = 1$, $6/(3 \cdot 3) = \frac{2}{3}$, $12/(6 \cdot 3) = \frac{2}{3}$: exactly the similarity table of Section 4.
- It is **symmetric**, and it must be, by the transpose rule of the next subsection: $(EE^\top)^\top = (E^\top)^\top E^\top = EE^\top$.

One product computed all the pairwise comparisons at once. The other order, $E^\top E$, is 4×4 : its entries are dot products between the word *columns*, measuring which words appear together. That is the pattern of $X^\top X$: for a data matrix X with one row per patient, $X^\top X$ is a small square table of dot products between features, and it has one row and column per feature no matter how many patients there are.

Example 6.8: Converting one column: a diagonal matrix on the right. Section 4 noted that a scalar cannot convert just one column of a data table. A diagonal matrix can. Take three patients' age (years) and fasting glucose (mg/dL), and the diagonal matrix with factors 1 and $\frac{1}{18}$:

$$X = \begin{bmatrix} 54 & 126 \\ 58 & 144 \\ 41 & 90 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{18} \end{bmatrix}, \quad XD = \begin{bmatrix} 54(1) + 126(0) & 54(0) + 126(\frac{1}{18}) \\ 58(1) + 144(0) & 58(0) + 144(\frac{1}{18}) \\ 41(1) + 90(0) & 41(0) + 90(\frac{1}{18}) \end{bmatrix} = \begin{bmatrix} 54 & 7 \\ 58 & 8 \\ 41 & 5 \end{bmatrix}.$$

The ages are untouched and the glucose column is now in mmol/L. By the column rule, column j of XD is X times column j of D , and column j of D has a single nonzero entry, so it just scales column j of X . Multiplying by a diagonal matrix on the **right** scales the **columns**; on the **left** it would scale the **rows** instead. (For example, the 3×3 diagonal matrix with factors 2, 1, 1 on the left of X would double patient 1's whole row.) Order matters, again.

Notebook U2-3_Matrices-2_MultiplicationAndPowers, §4, the bills computed

6.4 Rules, and the One That Fails

Many rules of ordinary multiplication carry over to matrices, as long as the shapes fit.

Rules that carry over from numbers

- **Associative:** $(AB)C = A(BC)$
- **Distributive:** $A(B + C) = AB + AC$

- **Identity:** $IA = AI = A$
- **Zero:** $0A = A0 = 0$
- **Transpose of a product:** $(AB)^\top = B^\top A^\top$, the order **reverses**

Associativity means a chain such as ABC can be written without parentheses, just as with numbers. The identity matrix plays the role of the number 1, and the zero matrix the role of 0. The transpose rule is the one surprise, and the shapes show why the order has to reverse: if A is $m \times n$ and B is $n \times p$, then AB is $m \times p$ and $(AB)^\top$ is $p \times m$. Now $B^\top A^\top$ is $(p \times n)(n \times m) = p \times m$, which fits, while $A^\top B^\top$ is $(n \times m)(p \times n)$, whose inner sizes do not even match in general.

The kit and supplier examples above were associativity at work: pricing the kits first, $(RK)\bar{x}$, or counting the supplies first, $R(K\bar{x})$, gave the same totals. The distributive rule can be checked just as concretely. With the worked example's $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$, and $C = \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix}$,

$$A(B + C) = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 6 & 6 \\ 9 & 7 \end{bmatrix} = \begin{bmatrix} 24 & 20 \\ 54 & 46 \end{bmatrix}, \quad AB + AC = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix} + \begin{bmatrix} 5 & -2 \\ 11 & -4 \end{bmatrix} = \begin{bmatrix} 24 & 20 \\ 54 & 46 \end{bmatrix}.$$

Watch out: the identity and the zero matrix change size to fit

The rule $IA = AI = A$ is exact for a square A . For a matrix that is not square, the identity on each side must have the size that makes the product exist. If C is 2×3 , then $I_2C = C$ and $CI_3 = C$, while I_3C is not even defined. With $C = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \end{bmatrix}$, both I_2C and CI_3 return C unchanged, as you can check row by row. The same goes for $0A = A0 = 0$: each 0 is the zero matrix of whatever shape fits.

Example 6.9: The transpose rule with shapes that are not square. Use the matrices of Example 6.2: A is 2×3 , B is 3×2 , and $AB = \begin{bmatrix} 2 & 7 \\ 20 & 2 \end{bmatrix}$, so $(AB)^\top = \begin{bmatrix} 2 & 2 \\ 7 & 20 \end{bmatrix}$. The rule says this should equal $B^\top A^\top$, which is $(2 \times 3)(3 \times 2) = 2 \times 2$:

$$B^\top A^\top = \begin{bmatrix} 2 & 0 & -1 \\ 1 & 3 & 5 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 2 & -1 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} 2+0+0 & 6+0-4 \\ 1+6+0 & 3-3+20 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 7 & 20 \end{bmatrix}. \checkmark$$

The unreversed product $A^\top B^\top$ *does* exist here, but it is $(3 \times 2)(2 \times 3) = 3 \times 3$, the wrong shape to equal a 2×2 matrix. Getting the order wrong does not always produce an error message; sometimes it quietly produces a different-sized answer.

What does not carry over: commutativity. In general,

$$AB \neq BA.$$

With the matrices from the worked example, AB was computed above, and reversing the order gives something else:

$$AB = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}, \quad BA = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 5 \cdot 1 + 6 \cdot 3 & 5 \cdot 2 + 6 \cdot 4 \\ 7 \cdot 1 + 8 \cdot 3 & 7 \cdot 2 + 8 \cdot 4 \end{bmatrix} = \begin{bmatrix} 23 & 34 \\ 31 & 46 \end{bmatrix}.$$

Why it fails: AB means “ B first, then A ,” and BA means the reverse, and doing two things in the opposite order usually lands somewhere else. Rotate a photo and then flip it, or flip it and then rotate it: the results differ.

The shapes can even differ. A $(2 \times 3)(3 \times 2)$ product is 2×2 , but $(3 \times 2)(2 \times 3)$ is 3×3 . And two 2×3 matrices cannot be multiplied at all, in either order, because the inner sizes are 3 and 2.

Example 6.10: Reversing the order, three ways. Each of these reverses a product computed earlier.

1. **Same shape, different entries.** For Example 6.1's matrices, $AB = \begin{bmatrix} -3 & 10 \\ 15 & -6 \end{bmatrix}$, while

$$BA = \begin{bmatrix} 1 & 4 \\ 5 & -2 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 2+0 & -1+12 \\ 10+0 & -5-6 \end{bmatrix} = \begin{bmatrix} 2 & 11 \\ 10 & -11 \end{bmatrix}.$$

Not one entry agrees.

2. **Different shapes.** For Example 6.2's matrices, AB was 2×2 , but BA is $(3 \times 2)(2 \times 3) = 3 \times 3$:

$$BA = \begin{bmatrix} 2 & 1 \\ 0 & 3 \\ -1 & 5 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \end{bmatrix} = \begin{bmatrix} 2+3 & 4-1 & 0+4 \\ 0+9 & 0-3 & 0+12 \\ -1+15 & -2-5 & 0+20 \end{bmatrix} = \begin{bmatrix} 5 & 3 & 4 \\ 9 & -3 & 12 \\ 14 & -7 & 20 \end{bmatrix}.$$

Nine entries against four: the two products are not even comparable.

3. **A pair that happens to commute.** Two diagonal matrices always commute, because each just scales coordinates and scaling in either order gives the same result:

$$\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 8 & 0 \\ 0 & 15 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}.$$

But as soon as nonzero entries appear off the diagonal, it usually fails:

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

The lesson of all three: commuting is the exception, and it has to be checked, never assumed.

"In general" means that some particular pairs do commute (the identity rule $IA = AI$ is one example), but you can never count on it. **The habit to build:** never swap the order of a matrix product without a reason.

Notebook U2-3_Matrices-2_MultiplicationAndPowers, §3, AB and BA compared

6.5 Exercise: Matrix Products

Exercise 6.1

Let

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \end{bmatrix}.$$

1. Compute AB and BA . Are they equal?
2. State the dimensions of AC and compute it.
3. Explain why CA does not exist.

4. Compute $(AB)^\top$ and $B^\top A^\top$ and confirm they match.

Solution to Exercise 6.1

1. Each entry is a row of the left matrix dotted with a column of the right one:

$$AB = \begin{bmatrix} 1(3) + 0(0) & 1(1) + 0(2) \\ 2(3) + 1(0) & 2(1) + 1(2) \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 6 & 4 \end{bmatrix},$$

$$BA = \begin{bmatrix} 3(1) + 1(2) & 3(0) + 1(1) \\ 0(1) + 2(2) & 0(0) + 2(1) \end{bmatrix} = \begin{bmatrix} 5 & 1 \\ 4 & 2 \end{bmatrix}.$$

Not equal. This is another instance of $AB \neq BA$: even two small, simple matrices usually fail to commute.

2. A is 2×2 and C is 2×3 . The inner sizes are 2 and 2, so the product exists, and the outer sizes say AC is 2×3 . Entry by entry:

$$AC = \begin{bmatrix} 1(1) + 0(0) & 1(2) + 0(1) & 1(0) + 0(3) \\ 2(1) + 1(0) & 2(2) + 1(1) & 2(0) + 1(3) \end{bmatrix} = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 5 & 3 \end{bmatrix}.$$

There is a quicker way to see row 2. Row 2 of A is $\begin{bmatrix} 2 & 1 \end{bmatrix}$, and every entry of row 2 of AC is “2 times the entry in row 1 of C , plus 1 times the entry in row 2 of C .” So row 2 of AC is $2 \begin{bmatrix} 1 & 2 & 0 \end{bmatrix} + 1 \begin{bmatrix} 0 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 2 & 5 & 3 \end{bmatrix}$, a combination of the rows of C .

3. CA would be $(2 \times 3)(2 \times 2)$. The inner dimensions are 3 and 2: a row of C has three entries and a column of A has two, so there is nothing to pair the third entry with. CA is **undefined**. Notice that AC exists while CA does not: the order of a product can decide whether it exists at all.

4. From part 1, $AB = \begin{bmatrix} 3 & 1 \\ 6 & 4 \end{bmatrix}$, so flipping it gives

$$(AB)^\top = \begin{bmatrix} 3 & 6 \\ 1 & 4 \end{bmatrix}.$$

For the other side, transpose each matrix and multiply in the reversed order:

$$B^\top A^\top = \begin{bmatrix} 3 & 0 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3(1) + 0(0) & 3(2) + 0(1) \\ 1(1) + 2(0) & 1(2) + 2(1) \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 1 & 4 \end{bmatrix}.$$

They match. The unreversed order would not:

$$A^\top B^\top = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix},$$

which is a different matrix. The order reverses under a transpose.

6.6 Extra Practice

Exercise 6.2

Let P be 3×2 , Q be 2×2 , and R be 2×3 . Which of PQ , QP , PR , RP , QR , RQ , and $P^\top P$ exist, and what shape is each one that does?

Solution to Exercise 6.2

Compare inner sizes each time.

- PQ : $(3 \times 2)(2 \times 2)$, inner $2 = 2$: a 3×2 matrix.
- QP : $(2 \times 2)(3 \times 2)$, inner $2 \neq 3$: **undefined**.
- PR : $(3 \times 2)(2 \times 3)$, inner $2 = 2$: a 3×3 matrix.
- RP : $(2 \times 3)(3 \times 2)$, inner $3 = 3$: a 2×2 matrix. (So PR and RP both exist but have different shapes.)
- QR : $(2 \times 2)(2 \times 3)$, inner $2 = 2$: a 2×3 matrix.
- RQ : $(2 \times 3)(2 \times 2)$, inner $3 \neq 2$: **undefined**.
- $P^\top P$: $P^\top$ is 2×3 , so $(2 \times 3)(3 \times 2)$: a square 2×2 matrix. A matrix times its own transpose, in either order, always exists and is always square.

Exercise 6.3

Let $\bar{u} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$ and $\bar{v} = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}$. Compute $\bar{u}^\top \bar{v}$ and $\bar{v} \bar{u}^\top$, and describe how the rows of $\bar{v} \bar{u}^\top$ are related to $\bar{u}$.

Solution to Exercise 6.3

$\bar{u}^\top \bar{v}$ is $(1 \times 3)(3 \times 1) = 1 \times 1$, the dot product:

$$\bar{u}^\top \bar{v} = 1(3) + (-1)(1) + 2(0) = 3 - 1 + 0 = 2.$$

$\bar{v} \bar{u}^\top$ is $(3 \times 1)(1 \times 3) = 3 \times 3$; entry (i, j) is $v_i u_j$:

$$\bar{v} \bar{u}^\top = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 3 & -3 & 6 \\ 1 & -1 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

Row i is v_i times the row $\bar{u}^\top$: row 1 is $3\bar{u}^\top$, row 2 is $\bar{u}^\top$ itself, and row 3 is $0\bar{u}^\top$. The same two vectors gave a single number one way and nine numbers the other way.

Exercise 6.4

In the college bill application, add a third scenario in which both colleges offer a 5% discount on the *current* rates. Write the 2×3 cost matrix and compute all three scenarios' bills in one product.

Solution to Exercise 6.4

A 5% discount multiplies each rate by 0.95: $0.95(60) = 57$ and $0.95(80) = 76$. The third column of the cost matrix is $\begin{bmatrix} 57 \\ 76 \end{bmatrix}$:

$$\begin{bmatrix} 10 & 7 \\ 11 & 4 \end{bmatrix} \begin{bmatrix} 60 & 66 & 57 \\ 80 & 88 & 76 \end{bmatrix} = \begin{bmatrix} 1160 & 1276 & 1102 \\ 980 & 1078 & 931 \end{bmatrix}.$$

The new column: $10(57) + 7(76) = 570 + 532 = 1102$ and $11(57) + 4(76) = 627 + 304 = 931$. The shapes $(2 \times 2)(2 \times 3) = 2 \times 3$ say it: two students by three scenarios. Adding a scenario added a column to the cost matrix and a column to the answer, and nothing else changed.

Exercise 6.5

Three patients received doses of 2, 3, and 5 mg. In regression the data matrix often gets a first column of 1s, so here $X = \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 1 & 5 \end{bmatrix}$. Compute $X^\top X$, say what each of its entries adds up, and check that it is symmetric.

Solution to Exercise 6.5

$X^\top$ is 2×3 and X is 3×2 , so $X^\top X$ is 2×2 . Its entries are dot products between the *columns* of X (column 1 is all 1s, column 2 is the doses):

$$X^\top X = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 5 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 1 & 5 \end{bmatrix} = \begin{bmatrix} 1+1+1 & 2+3+5 \\ 2+3+5 & 4+9+25 \end{bmatrix} = \begin{bmatrix} 3 & 10 \\ 10 & 38 \end{bmatrix}.$$

The top-left entry counts the patients (3), the off-diagonal entries add up the doses (10 mg in total), and the bottom-right entry adds up the squared doses (38). It is symmetric because both off-diagonal entries are the same dot product, column 1 with column 2, taken in either order; in general $(X^\top X)^\top = X^\top (X^\top)^\top = X^\top X$. With 1000 patients, X would be 1000×2 but $X^\top X$ would still be 2×2 : the summary that least squares works from, in four numbers.

Exercise 6.6

Show with numbers that $\begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}$ and $\begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}$ commute, and explain why any two 2×2 diagonal matrices do.

Solution to Exercise 6.6

In one order,

$$\begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} 2(3) + 0 & 0 + 0 \\ 0 + 0 & 0 + (-1)(4) \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ 0 & -4 \end{bmatrix},$$

and the other order gives the same matrix:

$$\begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 3(2) + 0 & 0 + 0 \\ 0 + 0 & 0 + 4(-1) \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ 0 & -4 \end{bmatrix}.$$

In general, every off-diagonal term in the product of two diagonal matrices has a 0 factor, so only the diagonal survives, and there the entries are $a_{11}b_{11}$ and $a_{22}b_{22}$ in one order and

$b_{11}a_{11}$ and $b_{22}a_{22}$ in the other. Ordinary numbers commute, so these agree. In the language of Section 5: scaling each coordinate by one factor and then another gives the same result in either order.

7 Matrix Powers and Networks

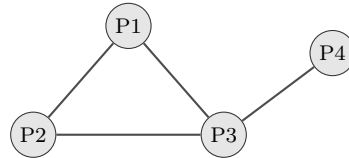
7.1 The Adjacency Matrix

Here is a **contact tracing** problem. In a waiting room, P1 sat with P2 and P3, P2 sat with P3, and P3 sat with P4. Suppose P1 later turns out to be infectious. Who sat with P1 is easy to read off the list. The harder question, and the one this section answers, is who could have been exposed *second-hand*, through someone who sat with P1.

Definition: Adjacency matrix

Draw each patient as a node and each shared sitting as a line. The **adjacency matrix** records the drawing: $a_{ij} = 1$ if i and j are directly connected, and $a_{ij} = 0$ if not.

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$



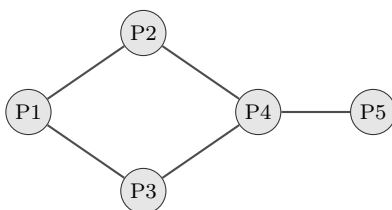
Row i lists who person i sat with. P1 sat with P2 and P3, so row 1 has 1s in columns 2 and 3: $\begin{bmatrix} 0 & 1 & 1 & 0 \end{bmatrix}$. The other rows are filled the same way.

Reading the matrix:

- It is **symmetric**, because sitting together is mutual: if P1 sat with P2 then P2 sat with P1, so $a_{12} = a_{21}$. The diagonal is all 0, since nobody is counted as a contact of themselves.
- Each **row sum** is a person's number of contacts, called their **degree**. The row sums are 2, 2, 3, 1, so P3, with three contacts, is the hub.

In words, an adjacency matrix is a yes/no table of “who is connected to whom,” with one row and one column per person. It turns a picture into numbers a computer can work with; a hospital’s contact log with hundreds of patients has no useful picture, but it has a perfectly good matrix.

Example 7.1: From a contact list to a matrix. Five clinic visitors are traced: P1 sat with P2 and P3, P2 with P4, P3 with P4, and P4 with P5. Build the adjacency matrix and find the degrees.



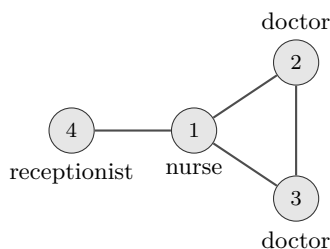
$$C = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

Each contact fills *two* entries, once in each person's row: the contact P2 with P4 puts a 1 at position (2, 4) and another at (4, 2). That is why the matrix is symmetric, and also why its entries add up to 10, twice the 5 contacts. The row sums give the degrees 2, 2, 2, 3, 1, so P4 is the hub here and P5, with a single contact, is on the edge of the network.

Example 7.2: From a matrix to a network. Going the other way, read the network off this adjacency matrix for four staff members (a nurse, two doctors, and a receptionist, numbered 1 to 4):

$$T = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}.$$

Because T is symmetric, every contact shows up twice, so read only the entries *above* the diagonal: $t_{12} = t_{13} = t_{14} = 1$ and $t_{23} = 1$, while $t_{24} = t_{34} = 0$. There are four contacts: 1 with 2, 1 with 3, 1 with 4, and 2 with 3. The nurse (degree 3) worked with everyone; the receptionist (degree 1) worked only with the nurse. Drawn, it is a triangle of nurse and doctors with the receptionist hanging off the nurse:



As a check, the entries of T add up to 8, twice the four contacts.

Watch out: what can and cannot be an adjacency matrix

For a network of mutual contacts like these, the adjacency matrix must be **square** (one row and one column per person), contain only 0s and 1s, be **symmetric**, and have 0s on the diagonal. So $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ is not one (it says 1 sat with 2 but 2 did not sit with 1), and neither is $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$ (it counts person 1 as their own contact). A matrix like the first one is fine for a network *with directions*, which the end of this section takes up.

Notebook U2-3_Matrices-2_MultiplicationAndPowers, §5, the network beside its matrix

7.2 Squaring the Matrix Counts Two-Step Routes

A **power** of a matrix is the matrix multiplied by itself: $A^2 = AA$, $A^3 = AAA$, and so on. This only makes sense for a square matrix, since A must have as many columns as rows for AA to fit, and an adjacency matrix is always square.

Example 7.3: Powers of small matrices. Before networks, three powers of ordinary matrices, to get the arithmetic in hand. Each one is nothing more than the row-by-column rule of Section 6 applied to a matrix and itself.

1. Square $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ by the row-by-column rule:

$$A^2 = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1+6 & 2+8 \\ 3+12 & 6+16 \end{bmatrix} = \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix}.$$

2. A diagonal matrix is easy to power, because each factor just scales its own coordinate again: each diagonal entry gets raised to the power, and the zeros stay zero.

$$\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}^3 = \begin{bmatrix} 2^3 & 0 \\ 0 & 3^3 \end{bmatrix} = \begin{bmatrix} 8 & 0 \\ 0 & 27 \end{bmatrix}.$$

3. Some matrices show a pattern. With $U = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$,

$$U^2 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad U^3 = U^2 U = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix},$$

and each further factor of U adds 1 to the top-right entry. Because matrix multiplication is associative, U^3 can be computed as $U^2 U$ or $U U^2$; the grouping does not matter.

Watch out: A^2 is not “square every entry”

In the first example, squaring each entry of A would give $\begin{bmatrix} 1 & 4 \\ 9 & 16 \end{bmatrix}$, which is *not* $A^2 = \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix}$. A matrix power is repeated matrix multiplication. The same trap is built into **numpy: `A**2`** squares each entry, while **`A @ A`** or **`np.linalg.matrix_power(A, 2)`** computes A^2 . (Diagonal matrices are the one case where the two happen to agree.)

Watch out: algebra shortcuts that need $AB = BA$

With numbers, $(a + b)^2 = a^2 + 2ab + b^2$. With matrices, multiply out carefully, keeping the order of every product:

$$(A + B)^2 = (A + B)(A + B) = A^2 + AB + BA + B^2.$$

The middle terms combine into $2AB$ only when $AB = BA$, which (Section 6) usually fails. For $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, the pair from Example 6.10 that does not commute,

$$(A + B)^2 = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}^2 = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}, \quad \text{but} \quad A^2 + 2AB + B^2 = \begin{bmatrix} 6 & 4 \\ 4 & 4 \end{bmatrix}.$$

For an adjacency matrix the product rule gives

$$(A^2)_{ij} = \sum_k a_{ik} a_{kj}.$$

Every entry of A is 0 or 1, so each term $a_{ik} a_{kj}$ is 1 exactly when i is connected to k **and** k is connected to j , and 0 otherwise. (A product of 0s and 1s behaves like the AND of Unit 1-1.) Each term that equals 1 is a route from i through k to j , so summing over k counts the intermediate people.

Squaring counts two-step routes

For an adjacency matrix, $(A^2)_{ij}$ is the **number of two-step routes from i to j** .

For the waiting room,

$$A^2 = \begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 3 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}.$$

Check one entry. Row 1 of A is $\begin{bmatrix} 0 & 1 & 1 & 0 \end{bmatrix}$ and column 4 is $\begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix}^T$, so

$$(A^2)_{14} = 0 \cdot 0 + 1 \cdot 0 + 1 \cdot 1 + 0 \cdot 0 = 1 :$$

exactly one route, P1 to P3 to P4. The only nonzero term is the one with $k = 3$, and P3 is the person in the middle.

The answer to the contact-tracing question. Row 1 of A^2 lists P1's second-hand exposures: **one route each** to P2 (via P3), to P3 (via P2), and to P4 (via P3). P4 never sat with P1, yet is reachable in two steps. P4 is exactly the person a “who sat with P1?” search would miss.

Two more features of A^2 are worth reading. The diagonal of A^2 is each person's degree, 2, 2, 3, 1: a two-step route from a person back to themselves goes out to a contact and straight back, one route per contact. And $(A^2)_{34} = 0$ because P4's only contact is P3 itself: a two-step route from P3 to P4 would need someone in the middle who sat with both, and nobody but P3 sat with P4.

Example 7.4: Second-hand exposure in the five-person network. Square the matrix C of Example 7.1. Row 1 of C^2 is row 1 of C , $\begin{bmatrix} 0 & 1 & 1 & 0 & 0 \end{bmatrix}$, dotted with each column of C . Since that row picks out entries 2 and 3, each dot product is “entry 2 plus entry 3 of the column,” that is, the entry in row 2 plus the entry in row 3. So row 1 of C^2 is row 2 of C plus row 3 of C :

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 & 2 & 0 \end{bmatrix}.$$

The whole square is

$$C^2 = \begin{bmatrix} 2 & 0 & 0 & 2 & 0 \\ 0 & 2 & 2 & 0 & 1 \\ 0 & 2 & 2 & 0 & 1 \\ 2 & 0 & 0 & 3 & 0 \\ 0 & 1 & 1 & 0 & 1 \end{bmatrix}.$$

Now read row 1, the index case P1.

- $(C^2)_{14} = 2$: **two** routes from P1 to P4, through P2 and through P3. P4 never met P1 but had two separate chances of second-hand exposure, a higher-risk contact than one route would suggest.
- $(C^2)_{11} = 2$: P1's degree, two “out and back” routes. These are not new exposures.
- $(C^2)_{12} = (C^2)_{13} = 0$: no two-step route reaches P2 or P3 (P2 and P3 never met each other). They are direct contacts, found in C itself, not in C^2 .
- $(C^2)_{15} = 0$: P5 is not within two steps of P1. Any exposure of P5 needs at least three steps, which the next subsection counts.

To list everyone within *two* steps of P1, add the one-step and two-step counts. Row 1 of $C + C^2$ is $\begin{bmatrix} 0 & 1 & 1 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 2 & 0 & 0 & 2 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 1 & 2 & 0 \end{bmatrix}$: apart from P1 itself, P2, P3, and P4 have nonzero entries, and P5 does not. Matrix addition and a matrix power together answer the tracer's real question.

7.3 Higher Powers, and Directed Networks

The same argument works for any power.

The general fact

$(A^n)_{ij}$ counts the **walks of length n** from i to j : sequences of n connections that start at i and end at j .

It follows from the same argument applied one step at a time, since $A^n = A \cdot A^{n-1}$. Writing out that product, $(A^n)_{ij} = \sum_k a_{ik} (A^{n-1})_{kj}$: a walk of length n from i to j is a first step from i to some neighbor k , followed by a walk of length $n - 1$ from k to j . Assuming A^{n-1} counts the shorter walks, A^n counts the longer ones, and we already know A^2 works.

Two things to keep straight:

- A walk may **revisit** people. For example, $(A^3)_{12}$ includes the walk P1 to P2 to P1 to P2. If you want routes that never repeat a person, powers overcount.
- The entries **grow fast** as n increases. How fast, and which entries dominate, is what **eigenvalues** measure in §2.6, and it is the mathematics underneath Google's PageRank.

Example 7.5: Three-step walks in the waiting room. Multiply A^2 by A once more:

$$A^3 = A^2 A = \begin{bmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 4 & 1 \\ 4 & 4 & 2 & 3 \\ 1 & 1 & 3 & 0 \end{bmatrix}.$$

Take $(A^3)_{12} = 3$ and list the three walks of length 3 from P1 to P2:

$$P1 \rightarrow P2 \rightarrow P1 \rightarrow P2, \quad P1 \rightarrow P2 \rightarrow P3 \rightarrow P2, \quad P1 \rightarrow P3 \rightarrow P1 \rightarrow P2.$$

Every one of them revisits somebody, which is the overcounting just described: as exposure routes they are no more worrying than the direct contact P1 to P2. And $(A^3)_{11} = 2$ counts the two ways around the triangle, P1 to P2 to P3 to P1 and the same loop in the opposite direction. A nonzero diagonal entry of A^3 says that person sits in a closed triangle of contacts. (In the five-person network of Example 7.1 every diagonal entry of C^3 is 0: nobody there is in a triangle.)

To see the growth, add up all the entries of A^n for the waiting room, the total number of walks of length n between anyone and anyone:

n	1	2	3	4	5	6
total walks of length n	8	18	38	84	180	394

The total roughly doubles with each step, and it passes four thousand by $n = 9$, in a network of only four people. That rate of growth is a property of the matrix, and it is exactly what §2.6 will measure.

Example 7.6: Three-step exposure in the five-person network. Row 1 of C^3 is row 1 of C^2 times C . Row 1 of C^2 is $\begin{bmatrix} 2 & 0 & 0 & 2 & 0 \end{bmatrix}$, so it picks 2 times row 1 of C plus 2 times row 4 of C :

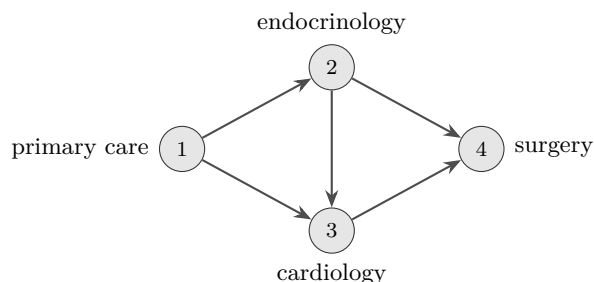
$$2 \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \end{bmatrix} + 2 \begin{bmatrix} 0 & 1 & 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 4 & 4 & 0 & 2 \end{bmatrix}.$$

So $(C^3)_{15} = 2$: P5 is reachable from P1 in three steps, by two walks, P1 to P2 to P4 to P5 and P1 to P3 to P4 to P5. In contact-tracing terms P5 is a *third-degree* contact, reachable only through the hub P4. The entries $(C^3)_{12} = (C^3)_{13} = 4$, on the other hand, are inflated by walks that double back (such as P1 to P2 to P1 to P2); P2 and P3 were direct contacts all along.

Directed networks. Some connections have a direction: page i links **to** page j , or person i follows person j , and the link need not be returned. For these, set $a_{ij} = 1$ for a link **from** i **to** j only. Then:

- A is no longer automatically **symmetric**, since a link from i to j does not imply one from j to i ;
- the row sums count **outgoing** links, and the column sums count **incoming** ones;
- $(A^2)_{ij}$ still counts the two-click routes from i to j , by exactly the same argument.

Example 7.7: A referral network. Patients move between four services by referral: primary care (1) refers to endocrinology (2) and cardiology (3); endocrinology refers to cardiology and surgery (4); cardiology refers to surgery. An arrow means “refers to.”



$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Row i lists where service i sends patients. Reading it:

- **Not symmetric:** $a_{12} = 1$ but $a_{21} = 0$. Endocrinology does not refer back to primary care.
- **Row sums** (referrals sent) are 2, 2, 1, 0; surgery refers nobody onward. **Column sums** (referrals received) are 0, 1, 2, 2; nobody refers into primary care.
- **Two-step pathways.** Squaring,

$$A^2 = \begin{bmatrix} 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

$(A^2)_{14} = 2$: two referral pathways lead from primary care to surgery in two steps, one through endocrinology and one through cardiology. $(A^2)_{13} = 1$: one two-step pathway to cardiology, through endocrinology.

- **Longer pathways die out.** A^3 has a single nonzero entry, $(A^3)_{14} = 1$, for the pathway primary care to endocrinology to cardiology to surgery, and A^4 is the zero matrix: no referral pathway is four steps long.

Why do the powers eventually vanish? Every arrow in this network points “forward,” from a lower number to a higher one, so no walk can ever return to where it started, and with only four services a walk cannot keep moving forward for four steps. In the waiting room, where contacts run both ways, the powers never vanish; they grow.

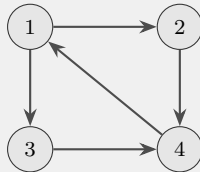
One more use for the transpose: $A^\top$ is the adjacency matrix of the same network with *every arrow reversed*, since $(A^\top)_{ij} = a_{ji}$ is 1 exactly when j refers to i . Its row sums, 0, 1, 2, 2, are the referrals each service *receives*: the column sums of A , as they must be.

The exercise that follows is a directed one.

7.4 Exercise: Two Clicks Away

Exercise 7.1

Four web pages link to each other as drawn. An arrow from i to j means page i links to page j .



Five links among four pages.

1. Write the adjacency matrix A , with $a_{ij} = 1$ for a link from i to j . Is it symmetric?
2. Compute **row 1** of A^2 . Which pages are exactly two clicks from page 1, and by how many routes?
3. Compute the entry $(A^2)_{41}$ and explain the answer in words.
4. Without computing, say which row of A^2 must equal row 2, and why.

Solution to Exercise 7.1

1. Row i lists where page i links **to**. Page 1 links to pages 2 and 3, pages 2 and 3 each link to page 4, and page 4 links to page 1:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}.$$

There are five 1s, one per link. A is **not symmetric**: $a_{12} = 1$ but $a_{21} = 0$, since page 2

does not link back to page 1.

2. Row 1 of A is $\begin{bmatrix} 0 & 1 & 1 & 0 \end{bmatrix}$. Dot it with each column of A in turn:

$$\text{column 1} = \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix}^T : 0 + 0 + 0 + 0 = 0,$$

$$\text{column 2} = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}^T : 0 + 0 + 0 + 0 = 0,$$

$$\text{column 3} = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}^T : 0 + 0 + 0 + 0 = 0,$$

$$\text{column 4} = \begin{bmatrix} 0 & 1 & 1 & 0 \end{bmatrix}^T : 0 + 1 + 1 + 0 = 2.$$

So row 1 of A^2 is $\begin{bmatrix} 0 & 0 & 0 & 2 \end{bmatrix}$. Only **page 4** is exactly two clicks from page 1, and by **two routes**: 1 to 2 to 4, and 1 to 3 to 4. The 2 counts routes, not pages.

3. $(A^2)_{41}$ is row 4 of A , $\begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}$, dotted with column 1 of A , $\begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix}^T$:

$$(A^2)_{41} = 1 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 + 0 \cdot 1 = 0.$$

There is **no two-click route** from page 4 to page 1. Page 4's only link goes to page 1, and the second click must then leave page 1 for page 2 or page 3, so you can never be on page 1 after exactly two clicks. (There is a one-click route from page 4 to page 1, but that is counted by A , not A^2 .)

4. **Row 3.** Rows 2 and 3 of A are identical, since each of those pages links only to page 4. Row i of A^2 is built from row i of A alone (dotted with the same columns of A), so identical rows of A give identical rows of A^2 : pages 2 and 3 have exactly the same two-click reach. Both rows are $\begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}$, the single route through page 4 to page 1.

7.5 Extra Practice

Exercise 7.2

An infection passes along a chain of four people: P1 met P2, P2 met P3, and P3 met P4, and nobody else met. Write the adjacency matrix A and its degrees, compute A^2 , and use A^3 's entry $(A^3)_{14}$ to say how P1 and P4 are connected.

Solution to Exercise 7.2

Each meeting fills two symmetric entries:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad \text{degrees } 1, 2, 2, 1.$$

Row i of A^2 adds up the rows of A belonging to i 's contacts. P1's only contact is P2, so row 1 of A^2 is row 2 of A ; P2's contacts are P1 and P3, so row 2 of A^2 is row 1 plus row 3 of A ; and

so on:

$$A^2 = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}.$$

The diagonal is the degrees, as always, and $(A^2)_{13} = 1$ is the route P1 to P2 to P3. There is no two-step route from P1 to P4, since $(A^2)_{14} = 0$. For three steps, $(A^3)_{14}$ is row 1 of A^2 , $\begin{bmatrix} 1 & 0 & 1 & 0 \end{bmatrix}$, dotted with column 4 of A , $\begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix}^T$, which gives 1: exactly one walk of length 3, P1 to P2 to P3 to P4. P4 is a third-degree contact of P1, reachable by a single chain.

Exercise 7.3

Three accounts follow one another: account 1 follows accounts 2 and 3, account 2 follows account 3, and account 3 follows account 1. With $a_{ij} = 1$ when i follows j :

1. Write A , and give each account's number of accounts followed and number of followers.
2. Compute A^2 and explain the entry $(A^2)_{11}$.

Solution to Exercise 7.3

1. Row i lists whom i follows:

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}.$$

The row sums, 2, 1, 1, count accounts followed; the column sums, 1, 1, 2, count followers. Account 3 has the most followers.

2. Row by column,

$$A^2 = \begin{bmatrix} 0+0+1 & 0+0+0 & 0+1+0 \\ 0+0+1 & 0+0+0 & 0+0+0 \\ 0+0+0 & 1+0+0 & 1+0+0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

$(A^2)_{11} = 1$ counts the one two-step chain from account 1 back to itself: 1 follows 3, and 3 follows 1. In a directed network a diagonal entry of A^2 is no longer the degree; it counts the *mutual* follows, since going out and straight back needs the link in both directions. Account 2 has $(A^2)_{22} = 0$: nobody it follows follows it back.

Exercise 7.4

For the web pages of the “Two Clicks Away” exercise, compute $(A^3)_{11}$ from row 1 of A^2 , and list the walks it counts.

Solution to Exercise 7.4

$A^3 = A^2A$, so $(A^3)_{11}$ is row 1 of A^2 , $\begin{bmatrix} 0 & 0 & 0 & 2 \end{bmatrix}$, dotted with column 1 of A , $\begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix}^T$:

$$(A^3)_{11} = 0 + 0 + 0 + 2 \cdot 1 = 2.$$

There are two three-click round trips from page 1: 1 to 2 to 4 to 1, and 1 to 3 to 4 to 1. The two two-click routes to page 4 each extend by the single link from page 4 back to page 1, which is exactly what the product computed: 2 routes to page 4, times 1 link onward.

Exercise 7.5

Compute $\begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}^4$. Then, using the pattern in the powers of $U = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ found earlier, predict U^4 and check it with one multiplication.

Solution to Exercise 7.5

A diagonal matrix raises each diagonal entry to the power: $2^4 = 16$ and $(-1)^4 = 1$, so

$$\begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}^4 = \begin{bmatrix} 16 & 0 \\ 0 & 1 \end{bmatrix}.$$

For U , the top-right entry went 1, 2, 3 for U, U^2, U^3 , so the prediction is $U^4 = \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix}$. Check with one more factor of U :

$$U^4 = U^3 U = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1+0 & 1+3 \\ 0+0 & 0+1 \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix}. \checkmark$$

8 Review

What we covered: the object and its arithmetic

1. “Is $\bar{b}$ a combination of these vectors?” is $A\bar{x} = \bar{b}$ with the vectors as the **columns** of A . A solution is the recipe; an impossible equation like $0 = 2$ means $\bar{b}$ is out of reach.
2. An $m \times n$ **matrix** is a grid with m rows and n columns; the entry a_{ij} is in row i , column j . Rows first, always. A column vector is an $n \times 1$ matrix.
3. A **data table** is a matrix: rows are records, columns are features, and both readings matter.
4. Know on sight the **zero**, **identity**, and **diagonal** matrices.
5. **Addition** and **scalar multiplication** are entry by entry and need matching shapes; an image is a matrix and these operations edit it.
6. The **transpose** swaps rows and columns, $(A^\top)_{ij} = a_{ji}$. A is **symmetric** when $A^\top = A$, and $A + A^\top$ always is.

What we covered: multiplication and powers

7. $A\bar{x}$ is the **linear combination of A 's columns** weighted by $\bar{x}$; computed row by row, each entry is a **dot product**. A system is now one equation, $A\bar{x} = \bar{b}$.
8. $I\bar{x} = \bar{x}$, and a **diagonal** matrix scales each coordinate by its own factor.

9. $(AB)_{ij}$ is **row i of A dotted with column j of B** ; the shapes must chain as $(m \times n)(n \times p) = m \times p$. Each column of AB is A times a column of B , so AB means “ B first, then A .”
10. Products are associative and distributive, $(AB)^\top = B^\top A^\top$, but $AB \neq BA$ in general.
11. For an **adjacency matrix**, row sums are degrees and $(A^n)_{ij}$ counts the **n -step walks** from i to j .

Notebooks: U2-3_Matrices-1_ArithmeticAndSpecial.ipynb
U2-3_Matrices-2_MultiplicationAndPowers.ipynb

Previous: Unit 2-2, **Vectors**.

Next: Unit 2-4, **Matrices as Transformations**.